

When it pays to be quick: dissociating control over task preparation and speed-accuracy trade-off in task switching

Bryant J. Jongkees, Michael T. Todd, Kevin Lloyd, Peter Dayan, Jonathan D. Cohen

Task switching paradigms offer an experimental window into the dynamics of cognitive flexibility. A key finding is that when switching from one task to another, participants exhibit a degradation in performance, even when given ample time to prepare for the switch. One hypothesis is that this “residual switch cost” stems from structural constraints on the cognitive system. However, an alternative hypothesis, which fits well with modern views on the costs of cognitive control, suggests the residual switch cost has a strategic origin. That is, timely preparation for a task switch incurs effort and/or opportunity costs, thus participants might choose not to do so, at least on some proportion of trials, unless adequately compensated. To demonstrate effort-sensitive, strategic control over task preparation, studies have aimed to offset the cost of preparation by offering monetary reward for responding by a deadline and observed reduced switch costs. However, rather than promoting timely preparation, such response deadlines might instead encourage speed-accuracy trade-offs following task preparation. A review of existing experiments involving reward and deadlines provides inconclusive results about how these factors influenced task switching performance. We therefore conducted a new experiment to provide a more sensitive test of the different hypotheses. Our data support a model in which response deadline for earning monetary reward affects speed-accuracy trade-off, while the preparatory process is unaffected. Accordingly, we argue that a reduction in switch cost due to reward-associated response deadlines does not necessarily provide evidence for strategic adjustment of task preparation. More generally, our approach demonstrates the utility of detailed modeling in conjunction with empirical testing in evaluating hypotheses concerning the dynamics of cognitive flexibility.

Keywords: task switching; failure to engage; cognitive modeling; dynamical systems
residual switch cost

1 Introduction

Task switching provides an empirical window into the dynamics of and constraints on cognitive control (Kiesel et al., 2010; Monsell, 2003; Musslick & Cohen, 2021). In particular, experiments have probed the factors underlying *switch costs*, a degradation in performance when switching between two simple tasks compared with repeating performance of either task. This was first reported by Jersild (1927), and Spector and Biederman (1976) later established boundary conditions under which switch costs occur. The modern task switching literature flourished in the 1990s, when Allport et al. (1994), Rogers and Monsell (1995), and Meiran (1996) established a standard paradigm for measuring switch costs, and demonstrated a *residual* switch cost: an impairment on switch trials that remains even given long preparatory intervals.

As we discuss below, there are several different interpretations of residual switch costs. In this article we focus on one: that these reflect the effort of maintaining preparation for an upcoming task, based on the observation that residual switch costs are reduced when this effort is compensated for by providing monetary reward for responding by a deadline. It has been argued that reductions in residual switch cost due to such factors (Nieuwenhuis & Monsell, 2002) provide support for effort-sensitive, strategic control over the task preparation process (De Jong, 2000; De Jong et al., 1999). Here, we offer empirical and theoretical support for an alternative: that reductions in switch cost due to a reward-associated response deadline instead reflect a speed-accuracy trade-off following task preparation. We provide evidence for this alternative explanation from model simulations and model fitting applied to new empirical data. While our findings are not meant to suggest that effort-sensitive, strategic control over task preparation does not exist, we suggest that more detailed experiments and analyses are required to demonstrate such effects firmly and specifically.

1.1 Structural vs Strategic Accounts

Interpretations of the residual switch cost can generally be placed into one of two broad categories. *Structural* interpretations presuppose that participants are always fully motivated to perform, and that the residual switch cost arises from (fixed) properties of the cognitive architecture. For example, Allport et al. (1994) interpreted the residual switch cost as evidence of “task set inertia” – a persistent activation of processes established during the previous task that

interferes with post-switch performance. Similarly, Altmann and Gray (2008) interpreted the residual switch cost in terms of proactive interference in episodic memory from previous task sets. Others (Meiran, 1996; Rogers & Monsell, 1995) have interpreted the residual switch cost in terms of an active preparatory process that cannot begin on a switch trial until after the first post-switch stimulus appears as it is required for some form of initialization.

An alternative *strategic* category of interpretations suggests that participants are able to fully prepare in advance of the new task but, at least on some portion of trials, may choose not to do so (De Jong, 2000; De Jong et al., 1999; Nieuwenhuis & Monsell, 2002). De Jong (2000) argued that reliably preparing for task switches is “optional” but effortful, consistent with traditional ideas about the effortful nature of cognitive control required to perform (or at least switch between) tasks (e.g., (Posner & Snyder, 2004), and anticipating more recent ideas about the costs of engaging cognitive control (Botvinick & Braver, 2015; Shenhav et al., 2013, 2017). De Jong (2000) further suggested that the residual switch cost arises from averaging performance from trials on which participants were fully prepared (and did not exhibit a switch cost) with those on which they did not expend the requisite effort to be prepared on time (and therefore exhibited a switch cost). This was supported by an analysis of reaction time (RT) distributions. In particular, it was observed that RT distributions on switch trials following a long-preparatory-interval were well fit by a mixture of a fast, “prepared” distribution (indexed by performance on repeat trials), and a slow, “unprepared” distribution (indexed by performance on switch trials following a short-preparatory-interval). To account for this finding, De Jong reasoned that participants might be able or willing to sustain the effort necessary for advance preparation only over brief periods of time. Consistent with this idea, it was observed that shorter trial blocks increased the estimated proportion of trials on which participants were prepared. This was taken as initial support for an effort-based origin of the residual switch cost.

Nieuwenhuis and Monsell (2002) later applied De Jong’s mixture model analysis to Rogers and Monsell’s (1995) original (non-incentivized) residual switch RT data, and then replicated that earlier experiment with two changes: shorter block lengths while keeping the total number of trials the same, and a monetary reward for fast, accurate responses, thereby effectively implementing a between-subjects manipulation of effort and its costs and benefits. They reported that both the old and new data were well fit by De Jong’s mixture model, and that the combination of short block length and monetary incentives increased the estimated proportion of prepared trials by 30%.

These data were offered as further support for the idea that effort-sensitive, strategic control over task preparation plays a role in the residual switch cost.

The conclusions described above fit well with recent theories on the expected value of control and the notion that effort associated with control leads people not to always exert it maximally (Botvinick & Braver, 2015; Shenhav et al., 2013, 2017). However, a wrinkle in the conclusions of De Jong (2000) and Nieuwenhuis and Monsell (2002) stems from the emphasis they placed on analysis of RT rather than accuracy. The potential concern is that encouraging fast performance might not make participants *prepare* earlier or more reliably, but instead (or perhaps as well) might make them *respond* faster once prepared. This would imply participants adopt a different speed-accuracy trade-off, leading them to make faster but less accurate responses. De Jong (2000) explicitly tested for but did not find a significant effect of block length (as a manipulation of sustained effort) on accuracy. However, those experiments included just eight to ten participants per group, and may therefore have been underpowered to detect the effect. Furthermore, a qualitative comparison of the data from Rogers and Monsell (1995) and Nieuwenhuis and Monsell (2002) does reveal higher error rates in the latter experiment where participants were rewarded for responding by a deadline (see Appendix A.1), despite the fact that participants were instructed to avoid errors. The higher error rates and faster RTs in Nieuwenhuis and Monsell's participants is consistent with a difference in the speed-accuracy trade-off between the experiments.

The possibility that participants might adjust their speed-accuracy trade-off following task preparation makes it difficult to interpret the effects of experimental manipulations like reward and response deadline on task switching. For example, consider the possibility that switch trial RTs are longer than repeat trial RTs due to a preparatory process that is strictly structural (i.e., insensitive to strategic control). Furthermore, imagine that participants must respond by a deadline, and that they are paid for doing so. In this situation, the optimal speed-accuracy trade-off may be different on switch as compared to repeat trials. This would be because the high baseline accuracy (typically above 90% in task switching experiments) could permit participants to sacrifice a small amount of accuracy on switch trials in order to compensate for the duration of the (fixed) preparatory process. This would help reduce the risk of the latter causing them to miss the deadline. Thus, even though the reward-associated deadline might not have affected the preparatory process itself, the switch trial RT would decrease more than the repeat trial RT when a reward-associated

deadline is present compared to when it is not, reducing the observed residual switch cost and creating the appearance of strategic control over task preparation.

The foregoing suggests that an experimental manipulation such as reward-associated response deadlines, even if meant to impact the preparatory process, might instead affect the decision making process. Conversely, it is also known that manipulations not involving monetary reward or deadlines — such as the probability of switches vs. repeats (Dreisbach & Haider, 2006; Musslick et al., 2018, 2019) — can impact preparatory processes. One way to help disentangle at least some of these confounds is to commit the processes involved to formal modeling, in which experimental manipulations (performance incentives and response deadlines) are taken into account, along with structural and dynamical factors (e.g., preparatory and decision making processes) in order to specify how strategic choices made by participants affects their behavior.

Taking this approach, in the next section, we describe a model that specifies distinct ways in which preparatory and decision processes may be sensitive to strategic choices. The model involves the coupling of two distinct processes, which we describe in more detail later: a *preparation* process, which determines how early and how intensively to prepare for an upcoming task, and a *decision* process, which involves a speed-accuracy trade-off. We assume that experimental manipulations (such as response deadlines and reward) can influence the speed-accuracy trade-off in the decision process, based on well-established literature on reward rate optimization through adjustment of speed and accuracy (e.g., (Balci et al., 2011; Bogacz et al., 2006). To clarify if strategic choices to obtain reward extend beyond the decision process and also influence task preparation, we obtain model fits in which the *timing* and *strength* of the preparation process is allowed to vary across experimental conditions and compare these with model fits in which these preparation processes are fixed across conditions. The former resembles De Jong’s (2000) hypothesis that participants strategically decide, based on experiment conditions, when timely preparation for a task switch is worthwhile. By contrast, the latter is more in line with structural interpretations of the residual switch cost where task preparation is insensitive to strategic control.

Following a description of the model in Section 2, in Section 3 we describe simulations using the model to explore how task switching performance changes as a function of task preparation and speed-accuracy trade-off, and compare these simulations to phenomena from previously reported task switching experiments. In Section 4, we report the results of a new

experiment designed to dissociate strategic control over preparatory and decision processes. The experiment is similar to that conducted by Nieuwenhuis and Monsell (2002), but uses a parametric, within-subject manipulation of the response deadline while keeping the magnitude of reward fixed. We find that the results of our experiment are best explained by the model where response deadline influences the speed-accuracy trade-off in the decision process, but *not* the timing or intensity of the preparation process. These findings suggest that, while strategic choices may influence the residual switch cost, this may be mediated by adjustment of processes other than task preparation. More generally, these findings highlight the value of detailed formal modeling of performance for interpreting the causal processes that determine performance in task switching experiments.

2 Non-Linear Dynamical System Modeling of Task Switching

2.1 Task Switching Experiment Structure

We consider a popular form of task switching experiment in which the participant switches between two tasks that involve the same input stimuli and the same output modalities, but different rules (Fig. 1, see also (Meiran, 1996)). For instance, participants are presented on each trial with a digit and must perform either a parity or a magnitude judgment: the parity task maps odd and even digits to left and right button press responses, respectively, while the magnitude task maps digits less than five to the left button and digits greater than five to the right button. In this design, stimuli can be either incongruent or congruent across tasks. For example, the stimulus ‘7’ requires a left response for the parity task and a right response for the magnitude task, and thus is response-incongruent; whereas the stimulus ‘3’ requires a left response for both tasks, and thus is response-congruent.

Participants perform the task on a sequence of hundreds of stimuli (trials), each of which is associated with a cue indicating which of the two tasks to perform on that trial. The cue is followed by a cue-stimulus interval (CSI), generally about one second in duration or less, after which the target stimulus appears and the participant must respond to the target according to the cued task. After participants respond, RT and accuracy are recorded, and the next trial begins. Here we focus on exogenous cueing (i.e., when there is an explicit cue indicating the task to be performed on each trial). This contrasts with the case in which tasks are performed in a predetermined sequence and participants must rely on a mental indication of which task to perform,

in which similar principles may apply, but processing may differ in ways that we do not address here.

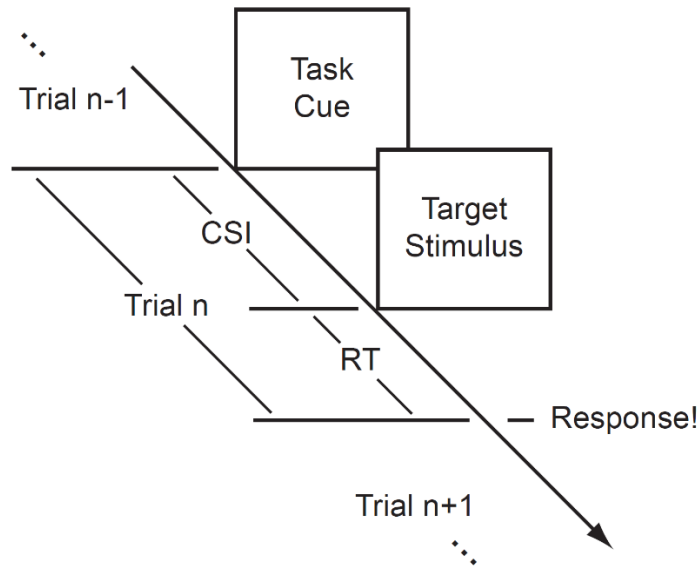


Figure 1: Task switching experiment trial structure.

2.2 Model Assumptions

We begin by providing an overview of key qualitative assumptions about task preparation and processing on which the model is based.

First, we assume that participants prepare for only one task at a time, given the possibility of incongruent stimuli. That is, as noted above, both tasks use the same stimuli but can map these to different responses; thus if participants attempted to prepare for both tasks simultaneously, the response for the cued task could be subject to interference from the response associated with the uncued task.

Second, we assume that mentally preparing a task takes time, for which there is considerable empirical evidence (for reviews, see (Kiesel et al., 2010; Vandierendonck et al., 2010). Note that even strategic accounts of the residual switch cost assume that the preparatory process takes time, differing from the structural accounts only on the issue of whether participants can decide when to initiate this time-consuming process (strategic account) or always do so at the earliest opportunity (structural account).

Third, we allow that decisions about mentally preparing a task — and then maintaining

that preparedness — can be effortful. For example, initiating preparation might be effortful due to the perturbation of working memory state associated with an update of its contents (e.g., (Kessler & Oberauer, 2014), the computations required to determine the contents and/or timing of the update (O'Reilly, 2006), as well as reflect the opportunity cost that preparing for and maintaining readiness for one task may carry given the first assumption, that preparation is limited to only a single task (see (Kurzban et al., 2013; Shenhav et al., 2017) for a more general consideration of such opportunity costs).

We also assume that task preparation is graded and that more intensive preparation requires more effort, but that this translates into responses that are both faster and more accurate (e.g., (Yeung & Monsell, 2003). This idea has been extended recently by Musslick et al. (2018, 2019), who showed it can be rational to adjust the intensity of task preparation to experimental conditions, and further showed that participants indeed make such adjustments. Thus, we extend the reasoning of De Jong (2000) and Nieuwenhuis and Monsell (2002) to consider not only *when* participants might prepare for a task, but also how *strong* or intensive their preparation is (as we explain in more detail in the next section).

Fourth, we assume that participants consider the consequences of their performance (i.e., speed and accuracy) when determining how to respond. This may be driven by explicit monetary rewards and response deadlines that are implemented in an attempt to influence preparation (e.g., (Nieuwenhuis & Monsell, 2002). However, we allow that it may also take account of other — sometimes implicit — experimental factors, such as the ubiquitous instruction to “respond as quickly and accurately as possible,” demand characteristics (e.g., the desire to impress the experimenter), and opportunity costs (i.e., the desire to finish the experiment as quickly as possible). We assume that all of these performance pressures may motivate participants to do their best to avoid incorrect responses and very long RTs.

Lastly, we assume that the speed-accuracy trade-off in the decision process is negotiated by a drift-diffusion model (DDM; (Ratcliff, 1978). This is a standard model family (e.g., (Bogacz et al., 2006; Gold & Shadlen, 2002) wherein noisy evidence is accumulated over time until one of two response thresholds is reached; at that moment the corresponding response is executed. We further assume that task preparation does not end abruptly once the DDM decision process begins but instead continues throughout evidence accumulation and translates directly to DDM drift; this implies a time-varying drift (e.g., (Liu et al., 2008; Shvartsman et al., 2015; Srivastava et al., 2017)

that is yoked to the progress of task preparation.

2.3 Model Description

We implement the assumptions outlined above in a model based on the neural network architecture first introduced by Cohen et al. (1990) to simulate task control in simple stimulus-response mapping tasks, combined with a non-linear dynamical system model of task preparation recently described by Musslick et al. (2018, 2019); see Appendix A.2 for detailed description of how the current model relates to these previous works). To summarize briefly, the model includes two non-linear processing pathways, one per task, that converge on a common response mechanism. Each is regulated by input from a control layer that contains units representing each task, the activity of which serves to increase the responsivity of units in the corresponding task pathway while suppressing activity in the other pathway. Accordingly, the activity of (and competition between) units in the control layer determine the dynamics of task preparation and performance.

The model contains an input layer, control layer, hidden layer, response layer, and a DDM that implements the decision process (Fig. 2A). Binary inputs T encode the currently-cued task (e.g., parity or magnitude) and are indexed by task $i \in \{1,2\}$. For simplicity, we assume that the stimulus inputs to the model, S , reflect the output of earlier stages of processing that extract the task relevant features along each dimension (i.e., parity and magnitude relative to 5) and, accordingly, encode these as binary values (i.e., representing odd/even for parity input units and lower/higher than five for magnitude inputs units), that are indexed by $i, j \in \{1,2\}$, where i is again the task and j is the stimulus feature.

The task inputs T project to control units C , which represent the strength of preparation for each task and are indexed by task $i, i^* \in \{1,2\}$, $i^* \neq i$. The control units are leaky competing accumulators (LCAs) (Usher & McClelland, 2001) and their activity is driven by their respective net inputs:

$$dnet_i = \frac{dt}{\tau_{lca}} [T_i - \lambda \cdot net_i - \beta \cdot C_{i^*}] \quad (1)$$

where T is the unit's corresponding task cue input, set to 1 if the task is cued and 0 otherwise; λ is the decay rate, reflecting passive leakage of activation; β is the inhibitory weight applied to the activity of the competing control unit i^* ; and τ_{lca} is the unit's time step size. The net input of each control unit is related to its activity state by a nonlinear (logistic) activation function:

$$C_i = \frac{1}{1+e^{-g \cdot net_i}} \quad (2)$$

where g is a gain parameter that amplifies input and constrains the minimum and maximum levels of activity (Fig. 2B). In the limit, control unit activity is constrained between 0 and 1.

The control units project to hidden units H , which represent the input features for the corresponding task (e.g., odd and even in the parity pathway; low and high in the magnitude pathway), and are indexed by task $i \in \{1,2\}$ and stimulus feature $j, j^* \in \{1,2\}, j^* \neq j$. The hidden units' respective net inputs are:

$$net_{i,j} = 1.3 \cdot (S_{i,j} - S_{i,j^*}) + 4 \cdot C_i - 4 \quad (3)$$

where $S_{i,j}$ is the unit's corresponding stimulus input (e.g., 'odd' in the parity pathway), set to 1 if the feature is present and 0 otherwise; S_{i,j^*} is the alternative stimulus input for that task (e.g., 'even' in the parity pathway); stimulus inputs are multiplied by 1.3, which represents a connection weight between input and hidden units that captures the strength of automatic stimulus processing and stimulus quality (Cohen et al., 1990), see Appendix for how this weight was determined); and C is the activity of the corresponding control unit. Each hidden unit is assumed to have a constant negative bias of -4, so that it is relatively inactive in the absence of input, and is activated only weakly by stimulus input unless activity from its control unit offsets this bias and thus increases its responsivity. The net input of each hidden unit is related to its activity state by the same logistic function as control units (see Eq. 2), with a gain parameter fixed at $g = 1$.

The hidden units project their activity to response units R , which represent the different responses (e.g., left and right button press) and are indexed by $k, k^* \in \{1,2\}, k^* \neq k$. The activity of response unit k is determined by the net input received from the hidden units:

$$net_k = 2 \cdot (H_{k,k} - H_{k,k^*} + H_{k^*,k} - H_{k^*,k^*}) \quad (4)$$

where the input from hidden units is multiplied by 2, which represents a connection weight between hidden and response units to capture the strength of stimulus-response mappings (Cohen et al., 1990), see Appendix for how this weight was determined). The net input of each response unit is again related to its activity state by the logistic function (see Eq. 2), with a gain parameter fixed at $g = 1$. We use the sign of the hidden unit inputs (positive or negative) to the response units to implement the response congruence of stimulus features across tasks. For example, in the case of the parity and magnitude tasks, hidden units 'odd' and 'low' provide excitatory input to the left response unit and inhibitory input to the right response unit, whereas the hidden units 'even'

and ‘high’ provide excitatory input to the right response unit and inhibitory input to the left response unit (see Appendix for a more thorough consideration of the role that the response units’ non-linearity plays in determining how congruent vs incongruent stimuli affect performance).

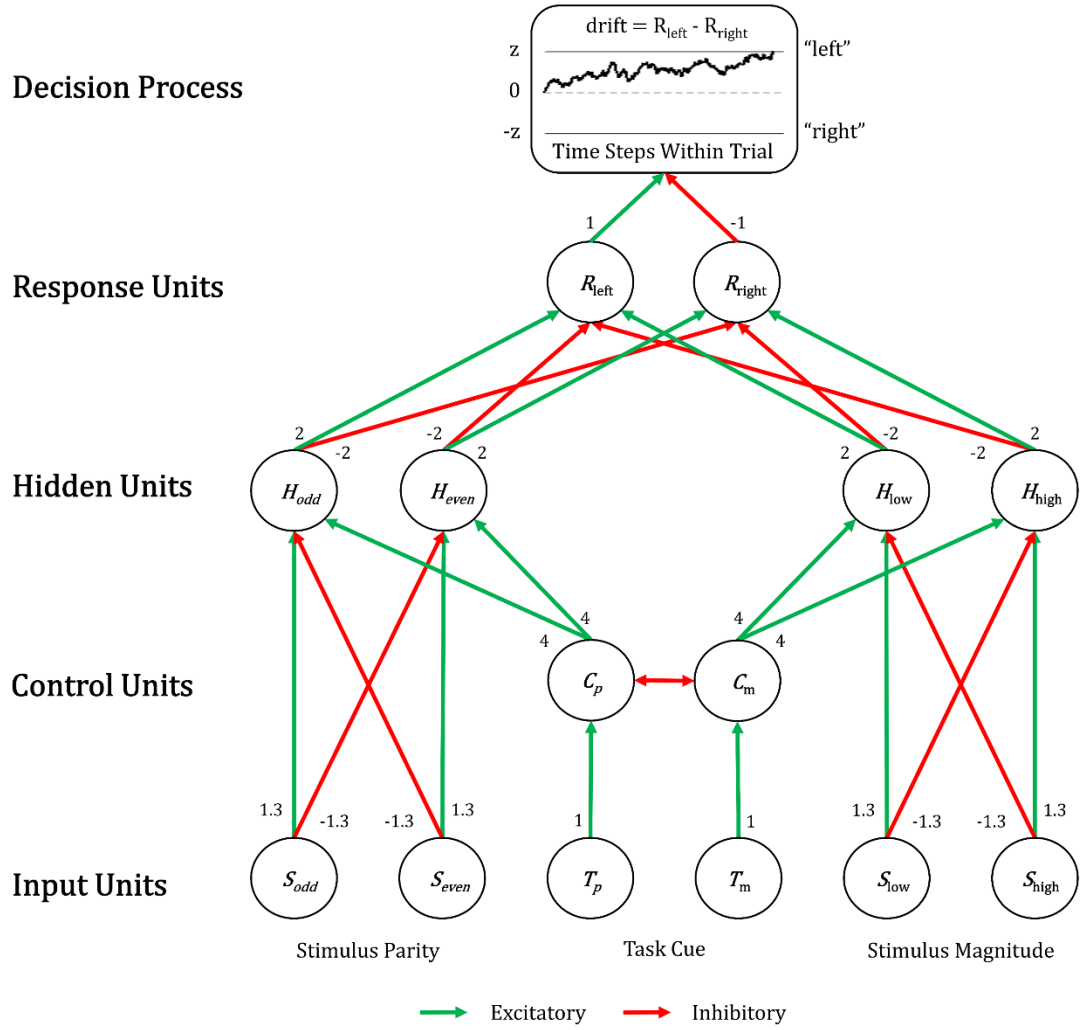
The response units provide input to a DDM that implements the decision process, which is assumed to determine the overt decision of the model. Evidence accumulation in the DDM starts unbiased and proceeds according to:

$$dx = \frac{dt}{\tau_{ddm}} [R_1 - R_2] + \varepsilon \frac{dt}{\tau_{ddm}}, x(0) = 0 \quad (5)$$

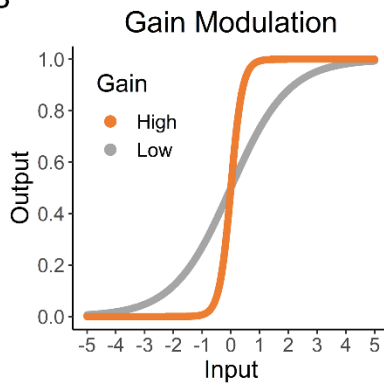
where the drift rate is the difference in activation of the response units, τ_{ddm} is the time step size of the DDM, and $\varepsilon \frac{dt}{\tau_{ddm}}$ represents Gaussian distributed white noise with mean 0 and variance ε .

Once evidence accumulation starts, the activity of the control units and the DDM decision process take timesteps in lockstep, meaning that post-stimulus changes in activation of the control units continuously propagate to the response units and translate into a time-varying drift rate for the DDM. This allows ongoing task preparation to dynamically influence the decision process. Evidence accumulation continues until decision threshold z or $-z$ is reached. Once a threshold is reached, the decision is recorded, and the next trial begins with the control unit activities persisting across trials.

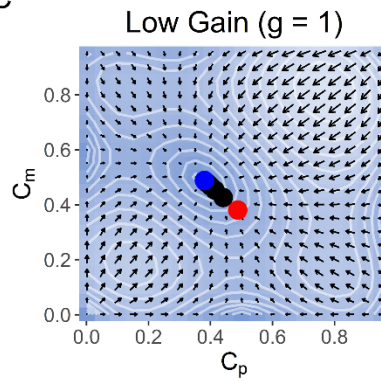
A



B



C



D

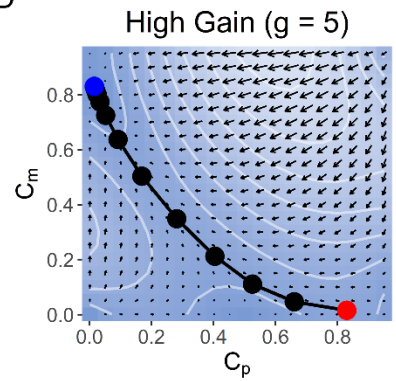


Figure 2: The non-linear dynamical system model of task switching. (A) Network architecture. The activity of task-representing control units (C_p = parity, C_m = magnitude) unfolds over the course of experiment trials, and is determined by excitatory task cue input, mutual inhibition,

decay, and gain modulation. The control units increase activity of the corresponding hidden units (H_i) that represent the four possible stimulus features. The hidden units provide input to two response units (R_i), which, in turn provide input to the DDM. On each trial, the DDM accumulates evidence based on the difference in activity of the two response units until a threshold is reached. Numbers next to arrows indicate the connection weights between units. (B) Effect of gain modulation on the nonlinear activation function. The activation function relates the net input of a unit to its activity state (see text for details). (C-D) Phase portraits for activity of the control units in response to inputs from T_i , under low gain (C) and high gain (D), showing trajectory from prior activation of one task (e.g., red dot for C_p) to the cued task (e.g., blue for C_m). Contour lines and arrows indicate the energy and shape of the attractor landscape after a task switch from one task to the other.

Use of LCAs for the control units implements the first two assumptions noted above: the inhibition and leak parameters ensure that the activity of one control unit will come to prevail over the other, thus causing the network to preferentially prepare and perform one task. Furthermore, these parameters, together with the gain parameter, determine the shape of the attractors associated with the prepared state for each task, which in turn influences the time required to transition between. Following previous work, we assume that inhibition and leak are fixed parameters of the LCA mechanism (see Appendix A.3 for more detail), but that gain may be subject to dynamic neuromodulatory regulation (Aston-Jones & Cohen, 2005; Servan-Schreiber et al., 1990). Changes in gain not only change the shape of the attractors, but also the trajectory of the transition between them, and thus the time it takes to switch between tasks. More specifically, increasing gain deepens the attractor associated with each control unit and increases the distance between them. Deepening an attractor corresponds to greater activity of the corresponding control unit (when fully prepared), which should improve performance on that task (e.g., reduce incongruence effects) by providing stronger “top down” support for processing in the corresponding pathway. However, at the same time, because of the increased depth of and distance between attractors, it takes longer to transition between them, thus increasing the switch cost (see Figure 2C-D). Such gain modulation of control units has been proposed previously (Musslick et al., 2018; Musslick & Cohen, 2021) as a mechanistic account for the stability-flexibility tradeoff in task switching settings (Goschke, 2000; Hommel, 2015).

Here we consider how CSI interacts with the activity dynamics of control units to provide an account of the empirically observed effects of CSI on switch costs. We do so by providing the cue to the task input units T_i at the start of the CSI, which it then processes (in the absence of any stimulus inputs) for a number of time steps corresponding to the CSI, thereby allowing it to “prepare” by activating the corresponding control unit C_i for the cued task. Thus, for short CSIs, the control units have less time to settle fully into the appropriate attractor state, resulting in less than maximal activity in the task-relevant hidden and response units, and thus impaired performance on switch trials. In contrast, on repeat trials, the control units have sufficient time (over the course of the previous trial) to settle fully into their attractor states, resulting in maximal activity of the task-relevant hidden and response unit and minimal activity in the task-irrelevant units, allowing for fast and accurate performance that is less sensitive to the length of CSIs. Gain modulation in the control units amplifies the effects described above. Specifically, high gain in the control units increases the time necessary to fully settle into the appropriate attractor state, resulting in stronger impairment of switch trial performance at shorter CSIs. Moreover, high gain enhances the contrast between activity in the task-relevant and task-irrelevant pathway once the control units have fully settled into the appropriate attractor state, thus augmenting performance on repeat trials.

These effects allow the model to capture the empirically observed effects of preparatory interval (CSI) on switch costs (e.g., (Nieuwenhuis & Monsell, 2002; Rogers & Monsell, 1995). Here, we use the model to compare these effects with those associated with the speed-accuracy tradeoff (i.e., changes in the threshold) of the decision process, and the extent to which manipulations of experimental factors (e.g. reward-associated response deadlines) are best explained by the dynamics of preparation versus changes in the speed-accuracy tradeoff.

3 Model Behavior

In this section, we begin by considering how task switching performance changes as a function of task preparation and speed-accuracy trade-off. This provides a reference for predicting how experimental factors (e.g., monetary reward and response deadlines) should affect switch costs as a function of the extent to which they impact task preparation versus the speed-accuracy trade-off in the decision process. We then draw comparisons between these model simulations and phenomena from key experiments described in the Introduction. Because we do not have access

to the raw data of these older experiments, we focus on the qualitative patterns of mean RT and accuracy across different experiment conditions.

3.1 Model Simulations

In the Introduction, we raised the possibility that experimental factors (such as monetary reward and response deadline) may impact not only task preparation, but also (or even instead) induce a change in the speed-accuracy trade-off in order to beat a response deadline. Furthermore, if switch trial RTs are longer than repeat trial RT due to a strictly structural (i.e., fixed) preparatory process, then favoring speed over accuracy would have a greater effect (i.e., decrease RT more) on switch compared to repeat trials. This would reduce the observed residual switch cost and create the appearance of strategic control over the preparatory process. However, the emphasis on speed would come at the cost of accuracy. Thus, while both increased task preparation and speed-accuracy trade-offs are expected to reduce switch costs in RT, they differ in the predicted change in accuracy.

To quantify this effect, we first simulated RT and accuracy for 10 sequences each consisting of 1024 trials with 50% task-switches and 50% response-incongruent stimuli. For each trial sequence we tested CSIs (number of time steps taken by the control units during the CSI) between 0 and 200 in increments of 20 while fixing all other parameters of the model (gain = 5, threshold = 0.16). We focus here on reporting performance on response-incongruent trials, as these most heavily depend on task preparation (the observed patterns show qualitatively similar, but less pronounced effects for response-congruent trials).

As the CSI increased, and the model had more time to process the cue and activate the corresponding control unit, performance on switch trials improved both in RT and accuracy. Performance on repeat trials was always high due to maximal task preparation (Fig. 3A, B). Consequently, switch costs in both RT and accuracy monotonically decreased as a function of CSI (Fig. 3C, D).

Second, we repeated the simulations above but varying the gain on the control units' activation function between 1 and 10 in steps of 1 while fixing all other parameters (CSI time steps = 50, threshold = 0.16). Overall, this revealed a monotonic increase in switch costs with increasing gain (Fig. 3I, J), consistent with previous work suggesting that increases in gain impair flexibility of processing (Musslick et al., 2018; Musslick & Cohen, 2021). Increasing gain also had the

expected effect of improving both RT and accuracy in repeat trials (Fig. 3G, H), consistent with the idea that deepening attractors of control representations augmented stability at the expense of flexibility. However, the effect of gain on performance on *switch* trials was more complex. Increases of gain above a moderate level (of approximately 5) produced the expected deterioration of performance on switch trials, consistent with its deleterious effect on flexibility. However, at lower levels of gain (below 5), gain had a *positive* effect on performance for switch trials as it did repeat trials (see left segment of curves in Figures 3G and H). This indicates that, not surprisingly, there is a minimum level of gain needed to exploit the effect of control. Importantly, this also reaffirms that, for a given set of task conditions, there is an optimal level of gain that optimizes the trade-off between stability and flexibility — an effect that has been explored in previous work (Musslick et al., 2018, 2019).

Third, we repeated the simulations to evaluate the effects of the speed-accuracy tradeoff on switch costs, by varying DDM threshold between 0.02 and 0.2 in steps of 0.02 while fixing all other parameters (CSI timesteps = 50, gain = 5). Confirming our key hypothesis, at lower thresholds (representing a stronger emphasis on speed over accuracy), RT and accuracy decreased both for switch and repeat trials but more strongly for switches (Fig. 3M, N). Consequently, at lower thresholds, switch costs monotonically decreased in RT but increased in accuracy (Fig. 3O, P).

Lastly, we examined how these changes in task preparation and speed-accuracy trade-off affect the overall reward rate and, more specifically, the probability of correct responding by a certain deadline, as is typically demanded in task switching experiments involving monetary reward (e.g., (Nieuwenhuis & Monsell, 2002)). Reward rate was calculated as mean accuracy divided by mean correct RT (Bogacz et al., 2006) and probability of beating the deadline was quantified by calculating the proportion of correct responses made before an arbitrary deadline (0.6s). Increasing CSI monotonically increased reward rate and proportion of correct responses before the deadline (Fig. 3E, F). Increasing gain similarly increased reward rate and facilitated beating the deadline, except for the highest levels of gain (due to degraded switch trial performance; Fig. 3K, L). Most relevant to our key hypothesis, the effect of threshold on beating the deadline was pronounced but not monotonic: to some extent, lowering threshold increased reward rate and proportion of correct responses before the deadline, although when threshold became too low the reward rate and proportion decreased due to a strong increase in error responses

(Fig. 3Q, R).

Taken together, these results support the hypothesis that the change in RT switch costs associated with earlier task preparation is qualitatively similar to the effects when time of preparation is fixed but speed is emphasized over accuracy (e.g., in order to meet a response deadline). The key difference is that accuracy on switch trials improves with earlier task preparation but deteriorates with an emphasis on speed over accuracy. As a result, the overall chance of successful responding before a deadline increases monotonically as task preparation starts earlier, but this can also be achieved, at least to some extent, by emphasizing speed over accuracy. In Section 4 we present data from a new empirical study aimed at dissociating these possibilities. First, however, we consider how the present model can address the widely observed residual switch cost effect.

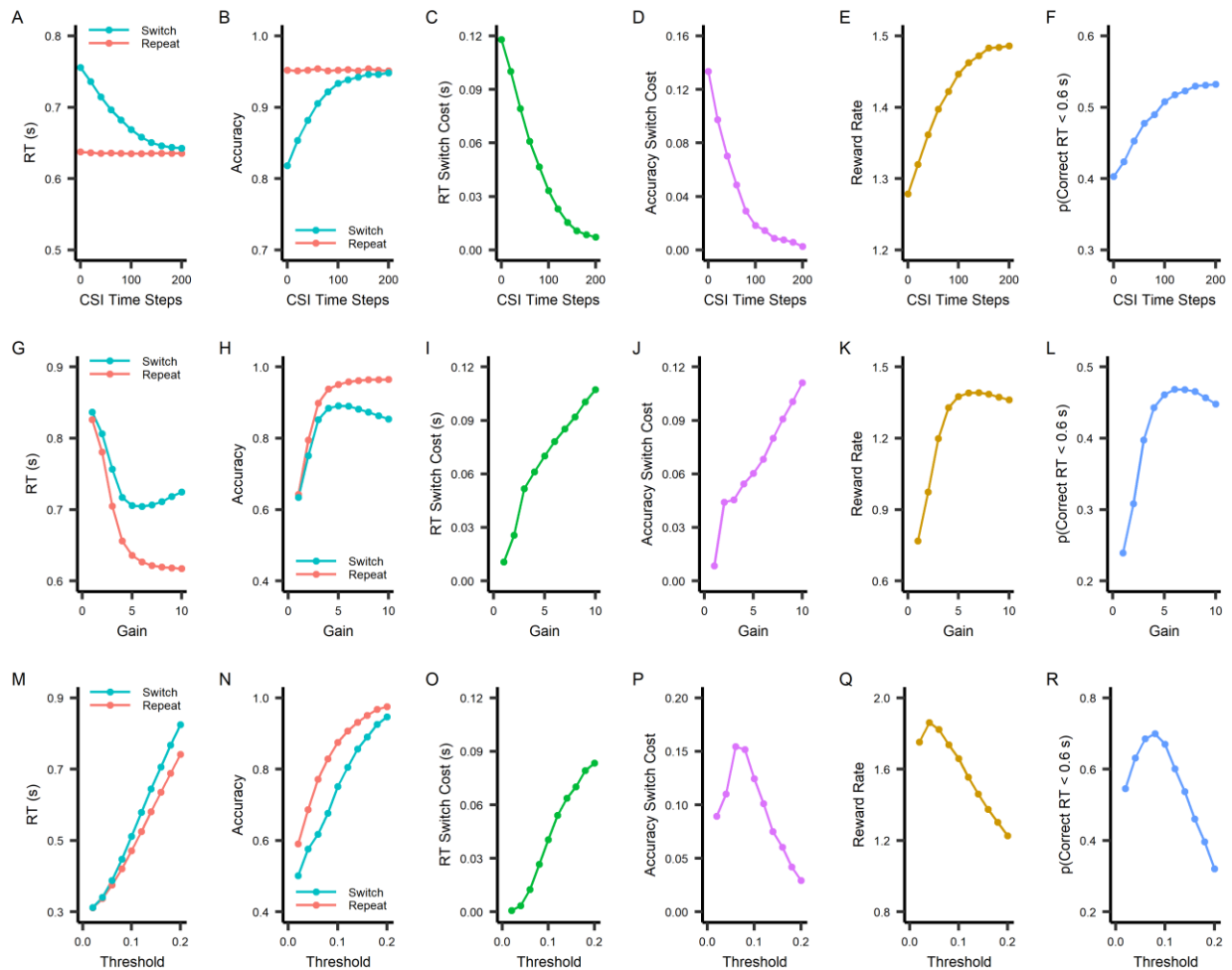


Figure 3: Simulation results. Mean RT and accuracy, switch cost in RT and accuracy, overall

reward rate, and proportion of correct responses before a deadline of 0.6 s as a function of CSI time steps (A-F), gain (G-L), and threshold (M-R). Switch cost is calculated as switch minus repeat in RT, and repeat minus switch in accuracy.

3.2 Residual Switch Costs (Rogers & Monsell, 1995)

In Experiment 3 of the classic study by Rogers and Monsell (1995), participants completed an alternating-runs task switching paradigm in which the task changed predictably every other trial (i.e., an AABBAABB sequence). Between trial blocks the response-stimulus interval was varied in the range from 0.15 to 1.2s; given the predictable task sequence, this interval served as a CSI. Figures 4A, B show the empirical data of Rogers and Monsell, which shares several qualitative patterns with our model simulations (see Fig. 3A-D). Rogers and Monsell found that RT and accuracy switch costs decreased as CSI increased. However, the RT switch cost asymptoted when the CSI passed 0.6 s, leaving a robust residual switch cost, while accuracy switch cost decreased without exhibiting a residual cost. Two things stand out in these data when compared to our model simulations.

First, there is a residual switch cost in Rogers and Monsell's empirical RT but not in our model-simulated RT. Our simulations suggest that sufficient task preparation should ultimately eliminate switch costs. As summarized in the Introduction, the observation that empirical switch costs asymptote and are not eliminated has been interpreted as participants not always using the CSI to maximally prepare for the task (strategic account; (De Jong, 2000; Nieuwenhuis & Monsell, 2002). In principle, this could be consistent with our model, if strategic choices limit or delay preparation on some trials. Alternatively, it has been suggested that this residual switch cost reflects a post-stimulus, exogenous component to task switching that cannot be completed during the CSI (structural account; (Meiran, 1996; Rogers & Monsell, 1995). However, the structural account has been questioned, since it predicts that long-CSI switch RT is not a *mixture* but rather a *shifted* version of the long-CSI repeat RT distribution. Yet, mixture modeling suggests that including such a shift does not improve model fit to the data (De Jong, 2000; Nieuwenhuis & Monsell, 2002). Thus, the residual switch cost in RT as observed by Rogers and Monsell appears to be consistent with a strategic constraint on task preparation during a long CSI, that could be consistent with our model.

Second, both Rogers and Monsell’s empirical accuracy and our model-simulated accuracy do not exhibit a residual switch cost, but for different reasons. In the empirical data, accuracy for task switches and repeats converges almost symmetrically; that is, repeat accuracy decreases with CSI in addition to switch accuracy improving. In contrast, in our model, repeat accuracy remains relatively constant across CSI. Nieuwenhuis and Monsell (2002) speculated that this pattern of repeat accuracy in the empirical data might reflect a loss of arousal or readiness with increasing CSI on repeat trials, and that participants compensate for this by adjusting their speed-accuracy trade-off such that their long-CSI RT remains fast at the cost of accuracy. Our model does not incorporate a mechanism that implements such loss of arousal or putative boredom. It would be interesting in future work to examine if changes in arousal, perhaps combined with a rational analysis of effort in prolonged task maintenance, can reproduce this pattern in accuracy.

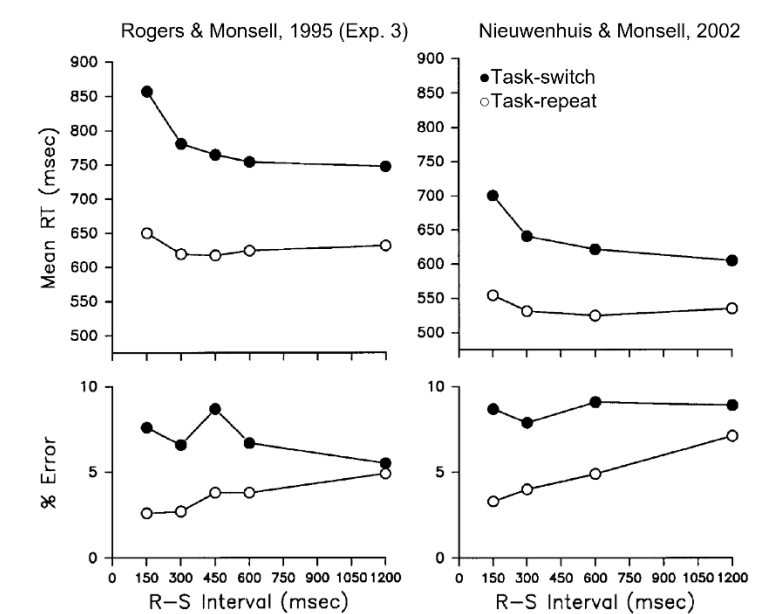


Figure 4: Data from Rogers and Monsell’s (1995) Experiment 3, as reported in Nieuwenhuis and Monsell (2002). RT (upper panels) and error rates (lower panels) are shown as a function of trial type (task switch versus repeat) and response-stimulus interval.

3.3 Failure to Engage (Nieuwenhuis & Monsell, 2002)

As noted above, De Jong (2000) argued that timely preparation on every long-CSI switch trial is “optional,” and that participants engage in preparation on some trials but fail to do so on

others. He supported this binary preparedness assumption¹ by showing that the cumulative distribution of RTs on long-CSI switch trials — and correspondingly the distribution of RT residual switch costs — is well fit by a mixture of the RT distribution for fast prepared trials and for slow unprepared trials. He constructed this mixture model from his data, using the RT distribution for long-CSI repeat trials to model the prepared component, and the distribution for short-CSI switch trials to model the unprepared component. The “Failure to Engage” hypothesis received subsequent support from Nieuwenhuis and Monsell (2002), who showed that a mixture model provided a similarly good fit to Rogers and Monsell’s (1995) data (Fig. 5A). Nieuwenhuis and Monsell further showed that the estimated proportion of prepared trials increased from .49 to .64 due to a combination of shorter block length and a monetary reward for responding by a deadline (Fig. 5B).

However, as suggested earlier, an alternative possibility is that the apparent increase in proportion of prepared trials could instead reflect a speed-accuracy trade-off that occurs when participants must respond within a pre-specified period, as they did in this experiment. To test the plausibility of this hypothesis, we conducted the following simulations. First, we simulated task-repeat RT after 200 CSI time steps to represent the ‘prepared’ distribution; task-switch RT after 20 CSI time steps to represent the ‘unprepared’ distribution; and task-switch RT with half of trials after 20 and the other half after 200 CSI time steps to represent the ‘residual’ (mixture) distribution. Thus, the preparation probability in this simulated residual distribution is 0.5, which matches almost exactly the 0.49 preparation probability inferred by Nieuwenhuis and Monsell (2002) for the unincentivized participants in Rogers and Monsell (1995). We then repeated these simulations for varying DDM thresholds (i.e., varying levels of speed-accuracy trade-off), and applied the mixture modelling analysis to the results of these simulations.

Figure 5C illustrates how the estimated proportion of prepared trials varied as a function of threshold, with all other factors being held constant. Two things stand out. First, the probability of prepared trials was somewhat underestimated compared to the 0.5 preparation probability used

¹ Intriguingly, at least one observation suggests the binary preparedness assumption of De Jong’s mixture model does not apply to the long-CSI switch RT distribution. As explained by Brown et al. (2006), the mixture model makes the strong prediction that if the probability distribution functions for prepared and unprepared RTs cross at some point, then all mixtures from those two must also cross at the same point. However, Brown et al. showed this was not true in empirical data, and while they demonstrated that an extended mixture model with additional parameters could accommodate the crossing point prediction, it casts further doubt on the all-or-none, binary preparedness assumption of De Jong’s theory.

to simulate the data. Second, the effect of threshold on estimated preparation probability was clearly non-monotonic. From low to moderate thresholds, the preparation probability increased, whereas from moderate to high thresholds the probability decreased again.

These simulation results suggest that Rogers and Monsell's (1995) unincentivized participants could have had relatively high thresholds and thus lower preparation probabilities (Fig. 5C, right section of curve) than the incentivized participants of Nieuwenhuis and Monsell (Fig. 5C, middle section of curve). This would be consistent with the observation that Rogers and Monsell's participants were both slower and more accurate. However, the simulation results also suggest that even relatively large differences in threshold produce only modest differences in preparation probability, and certainly not the 0.16 probability increase reported by Nieuwenhuis and Monsell.

In this regard it is noteworthy that Nieuwenhuis and Monsell implemented *two* changes compared to Rogers and Monsell's experimental design. Not only did they offer monetary reward contingent on beating a deadline, but they also incorporated shorter block lengths (16 vs 48 trials per block) while keeping the same overall number of trials. This difference may be important, as De Jong (2000) showed that a between-subjects manipulation of block length (12 vs 96 trials) produced a large difference in preparation probability (0.80 vs 0.57, respectively). This raises the possibility that the increased preparation probability in Nieuwenhuis and Monsell's incentivized participants is in large part due to shorter block length. This would be consistent with Nieuwenhuis and Monsell's own remark that the combination of shorter block length and monetary incentives only produced a 0.16 increase in preparation probability—perhaps because these two changes had opposite effects. However, without modelling the raw data from Rogers and Monsell (1995) and Nieuwenhuis and Monsell (2002), it remains difficult to determine if and to what extent their findings were confounded by a speed-accuracy trade-off. Nevertheless, it is clear that mixture modeling on its own cannot fully disentangle the effects of task preparation and speed-accuracy trade-offs. Accordingly, in the next section we report a new experiment designed to discriminate these different factors.

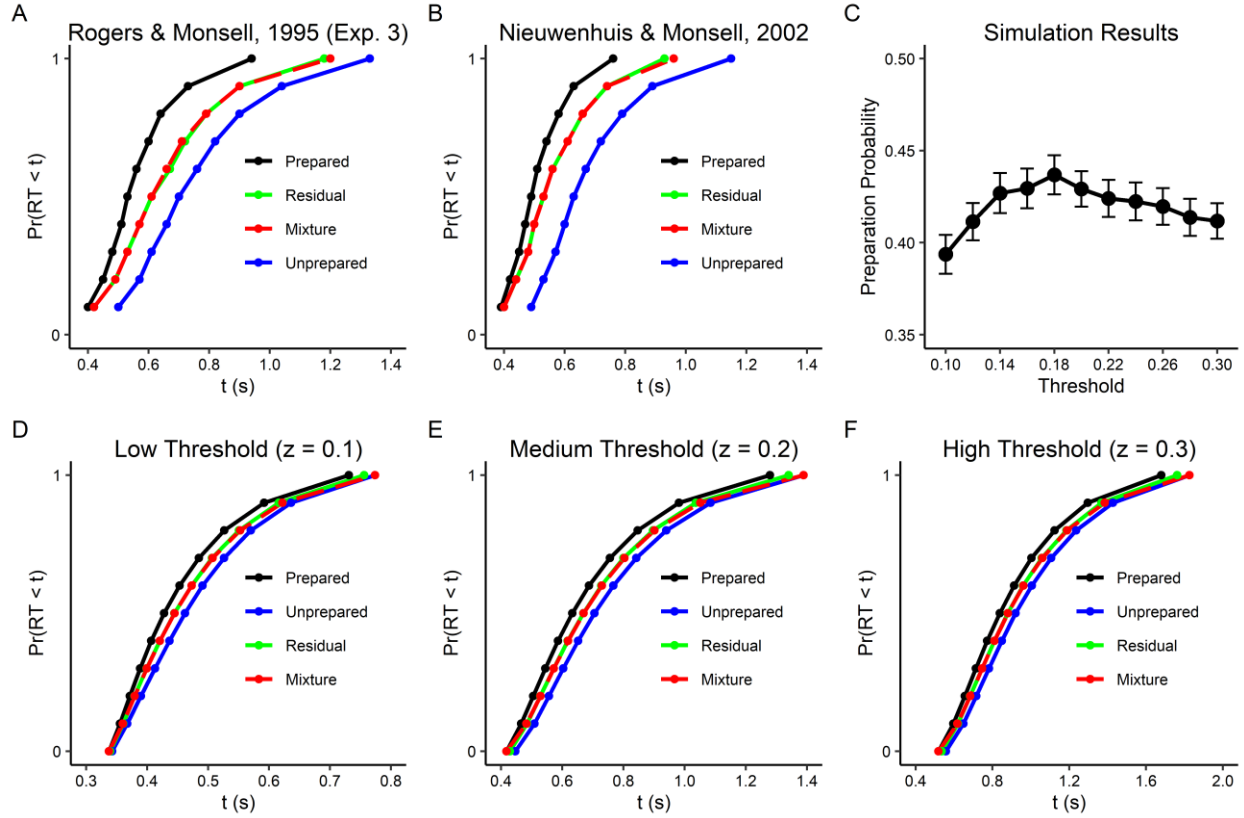


Figure 5: Mixture modelling of the Failure to Engage hypothesis. (A) Data from Rogers and Monsell (1995), as re-analyzed by Nieuwenhuis and Monsell (2002). (B) Data from Nieuwenhuis and Monsell (2002). (C) Mixture modeling of model simulated data with varying DDM threshold, error bars indicate ± 1 standard error of the mean. The horizontal axis indicates observed RT (A-B) or simulated RT (D-F) and the vertical axis indicates the cumulative RT distribution. The ‘residual’ long-CSI switch distribution (green line) is a mixture (red dashed line) of ‘prepared’ long-CSI repeat (black line) and ‘unprepared’ short-CSI switch (blue line) distributions, as shown by the overlapping red and green lines.

3.4 Summary of Basic Model Behavior

We presented model simulations showing how task switching performance changes as task preparation increases and speed is traded off against accuracy. We also show that our model captures many patterns observed in empirical data, and that mixture modelling of the residual switch cost provides inconclusive evidence regarding the factors that may influence performance in task switching experiments. In the section that follows, we use the model described above to examine the relative contribution of preparation versus speed-accuracy tradeoffs in a new

empirical study using a deadlined task switching paradigm designed to address this question.

4 Testing the Model: Deadlined Task Switching Paradigm

As discussed in the Introduction, the traditional approach to establishing an effort-sensitive, strategic origin of the residual switch cost has been through monetary reward. However, the design of previous experiments does not provide clear evidence for how reward, and response deadlines for earning this reward, influenced task switching performance. One of the first experiments probing the residual switch cost (Rogers & Monsell, 1995) offered reward that was implicit, constant, and guaranteed – in the sense that participants were rewarded for their participation in the experiment but not their level of performance. As such, this was unlikely to incentivize participants to expend the effort necessary for early and reliable task preparation.

One approach, then, would be to manipulate the magnitude of reward experimentally, based on the assumption that larger reward should offset the cost of task preparation more and thereby decrease, and in the limit abolish, the residual switch cost. However, evidence supporting this hypothesis is mixed. Savine and colleagues (2010) reportedly eliminated the residual switch cost by employing a monetary reward without actually specifying to participants the amount; this may have led participants to make optimistic assumptions about actual payoff magnitude. However, Kleinsorge and Rinkenauer's Experiment 1 (2012) found that RT switch costs were eliminated with a magnitude of monetary incentive that was both effectively signaled to participants (i.e., they could see how their performance-contingent bonus incremented over time) and rather low (the average bonus was approximately \$2, in addition to a fixed amount of about \$12, for a session of approximately 1 hour). A caveat in their experiment, however, was having only 8 participants per group, meaning their experiment might have been underpowered. A more statistically powered study by da Silva Castanheira and colleagues (2021) included 70 participants for whom they manipulated reward magnitude (\$0, \$0.01, and \$0.10) in a within-subjects design. They observed that, while gaining some reward led to reduced switch costs, there were no switch cost differences between low and high reward. Notably, however, their study did not provide preparatory intervals (i.e., CSI was 0 s) and thus cannot address the *timing* of advance preparation. One study that did involve a manipulation of reward magnitude while varying CSI was the classic experiment by Nieuwenhuis and Monsell (2002). They asked participants to continuously try to

reduce their overall RT across blocks while avoiding errors, and paid more for larger reductions in RT. However, this instruction made larger reward magnitude (meant to offset the cost of preparation) inextricably dependent on increasingly strict response deadlines (which likely encourage different speed-accuracy trade-offs). As such, it is difficult to disentangle the role that reward magnitude and response deadline had on their observed reduction in residual switch cost.

4.1 Deadlined Task Switching Experiment

To avoid these issues, we took an alternative approach to probing the strategic origin of residual switch costs. We designed a “deadlined” task switching paradigm that parametrically manipulated the time allowed to respond once the stimulus appeared. In this paradigm, participants earned fixed monetary rewards for correct responses made before the deadline. Correct responses made after the deadline earned nothing, and error responses made at any time incurred a fixed monetary penalty. The rationale was that changing the deadline while keeping all other factors constant would render a given RT and accuracy distribution more or less likely to yield monetary rewards, thus changing the expected pay-off for a given speed-accuracy trade-off. Using our model, we can capture the extent to which participants adjust their speed-accuracy trade-off to the deadline through the DDM’s threshold parameter. At the same time, we can capture if and how participants adjust the timing and intensity of their task preparation through the control units’ CSI time step and gain parameters. This allows us to examine if strategic choices, made to beat a deadline in order to earn reward, extend beyond adjustment of speed versus accuracy and also influence task preparation. As illustrated by our model simulations in the previous section (Fig. 3), both increasing task preparation and emphasizing speed over accuracy can increase the probability of correct responding by a deadline. However, the two strategies differ in their predictions of the extent to which response accuracy is sacrificed when trying to maximize the chance of beating the deadline.

Note that, the model described above makes two qualitative predictions, regardless of the strategy participants use to beat the deadline: an overall U-shaped relationship between mean RT and deadline, and an interaction of this with trial type (viz., switch vs. repeat, see Fig. 3). Regarding the U-shaped relationship: when the deadline is very short, there is little chance of integrating information required to make a correct response and earn a reward, along with the risk of penalty for responding quickly but incorrectly; thus, it makes sense to simply ignore the deadline and

integrate information in order to respond correctly in order to avoid a penalty. For similar reasons, there is no incentive to prepare early, since the deadline will be missed anyhow. At the other extreme, for sufficiently long deadlines, it should pay off to use the time to integrate as much information as possible in order to maximize the chance of earning a reward; furthermore, if the deadline is long enough, it will also afford time to prepare, thus obviating the need (and effort) to prepare in advance of the stimulus. Thus, together, longer integration time and no early preparation should result in longer RTs at the shortest and longest deadlines. In contrast, intermediate deadlines may afford enough time for integration to produce a correct response, providing that preparation occurred beforehand. Thus, participants may choose to prepare in advance and then choose a speed-accuracy trade-off that is just fast enough to beat the deadline while maximizing integration time, yielding RTs that are faster than for either the shortest or longest deadlines. Finally, this U-shaped relationship should interact with trial type, since switch trials are generally slower, exacerbating the effects described above.

The patterns of performance outlined above, reflecting the possibility that participants may adjust either or both task preparation and speed-accuracy trade-off in response to time available to respond, makes it difficult to qualitatively dissociate the effects of each, and thereby determine which may actually be used by participants. To address this, we carried out a quantitative model comparison, in which we fit the empirical RT and accuracy data of our deadlined task switching experiment using versions of our model in which task preparation was or was not allowed to vary across deadline conditions. We found that the version with fixed task preparation provided a better fit, suggesting that adjustments in the speed-accuracy tradeoff were sufficient to account for adjustment of behavior in response to the deadline manipulation.

4.1.1 Experiment Methods

Participants. 31 participants (20 female; mean age 20.4 years) were recruited from the Princeton University community. Participants were paid a base rate of \$12 for the 50-minute experiment, plus a performance-contingent bonus amount that was typically around \$10. Although bonuses could be negative (i.e., in the case of unusually poor performance), participants could not earn less than \$12 for the session. The experiment was approved by the Institutional Review Board of Princeton University.

Materials & Apparatus. Stimuli were presented and responses were recorded using Apple

iMac computers with 24-inch displays, running Matlab (The MathWorks Inc., Nattick, MA) with Psychophysics Toolbox (Brainard, 1997). Responses were made using the '<' and '>' keys on a standard 101-key keyboard.

Procedure. Participants performed parity/magnitude task switching. That is, on each trial they were cued which task to perform, and then presented with a single digit, excluding five, that they were required to judge with respect to whether it was odd or even (parity task), or lower or higher than five (magnitude task). “Odd” and “lower than five” mapped to the '<' response key; “even” and “higher than five” mapped to the '>' response key. Participants were given instructions and practice to ensure that they understood the experiment. They then progressed through eight experimental blocks with an unpredictable order of repeat and switch trials, and blocks separated by self-paced breaks with performance feedback.

Within each experimental block, the deadline was fixed at one of four levels (0, 0.6, 1.0, 1.4 s). Deadline order was randomized across participants, but constrained so that, within participant, deadline order was the same across the first four blocks as across the last four blocks, with each deadline sampled once in each set of four blocks. Each block contained 187 trials. Each trial began with a 900 ms CSI, which presumably provided participants sufficient time to prepare for a task switch (Rogers & Monsell, 1995), and thus probe residual switch costs. After each trial, the task had a 0.5 probability of switching. The specific task to be performed was signaled by a task cue (“OD-EV” for the parity task and “LO-HI” for the magnitude task) that was presented throughout the CSI. The target appeared at the end of the CSI. After the participant responded to the target, a “reward thermometer” appeared for 200 ms to indicate graphically whether the participant had earned or lost any money for the trial. Then the next trial began.

The reward for correct, on time responses, and the penalty for error responses were both fixed at \$0.01. Thus, for a correct response made before the deadline, participants earned \$0.01; for a correct response made after the deadline, participants earned \$0; and for an incorrect response made at any time, participants lost \$0.01.

Preprocessing & Analysis. For both RT and accuracy analyses, we excluded the first trial of each block. For RT, we additionally excluded very fast (less than 0.2 s) and very slow (more than 3 s) responses, followed by +/- 2.5 standard deviation trimming (Van Selst & Jolicoeur, 1994) as a function of participant, trial type (switch vs repeat), congruence (incongruent vs congruent), and deadline (0 vs 0.6 vs 1 vs 1.4 s). This excluded 0.97% of trials. Finally, we submitted mean

RT and accuracy to separate repeated-measures 2x2x4 ANOVAs with the within-subjects factors: trial type (switch vs. repeat), congruence (congruent vs. congruent), and deadline (0, 0.6, 1, and 1.4). Where the assumption of sphericity was violated we applied the Greenhouse-Geiser correction (Greenhouse & Geisser, 1959).

4.1.2 Behavioral Results

Experiment results are shown in Figure 6, presented both by trial type (Fig. 6A, B) and in terms of switch cost (i.e., difference between the trial types; Fig. 6C, D). ANOVA-based analyses are presented first, followed by model parameter estimates and Bayesian model selection below. For RT (Fig. 6A, C), there were significant main effects of trial type, $F(1,30) = 34.33$, $p < 0.001$, $\eta_p^2 = 0.534$, and deadline, $F(2.00,59.99) = 15.79$, $p < 0.001$, $\eta_p^2 = 0.345$. The predicted interaction was also significant, $F(3,90) = 4.33$, $p = 0.007$, $\eta_p^2 = 0.126$. Given the qualitatively U-shaped model predictions and the approximately regular progression of the levels of the deadline manipulation, we tested quadratic trends using polynomial contrasts. The main quadratic trend over deadlines was significant, $F(1,30) = 18.04$, $p < 0.001$, $\eta_p^2 = 0.376$, as was the interaction with trial type, $F(1,30) = 7.47$, $p = 0.010$, $\eta_p^2 = 0.199$, confirming that the quadratic trend across deadline was significantly stronger for switch than for repeat trials. Furthermore, there was a significant main effect of congruence ($F(1,30) = 69.68$, $p < 0.001$, $\eta_p^2 = 0.699$), indicating slower RT on incongruent than congruent trials (0.639 vs 0.592 s). There was also a marginally significant interaction between congruence and trial type ($F(1,30) = 3.13$, $p = 0.087$, $\eta_p^2 = 0.094$), indicating a trend towards larger switch costs on incongruent than congruent trials (0.072 vs 0.062 s). There were no interactions involving congruence and deadline ($ps \geq 0.488$).

For accuracy (Fig. 6B, D), an identical ANOVA revealed a main effect of deadline, ($F(2.35,70.62) = 5.00$, $p = 0.004$, $\eta_p^2 = 0.154$), and a significant quadratic trend over deadlines, $F(1,30) = 6.27$, $p = 0.018$, $\eta_p^2 = 0.173$. There was also a significant main effect of congruence ($F(1,30) = 76.70$, $p < 0.001$, $\eta_p^2 = 0.719$), indicating lower accuracy on incongruent than congruent trials (0.90 vs 0.98). No other effects were significant ($ps \geq 0.105$).

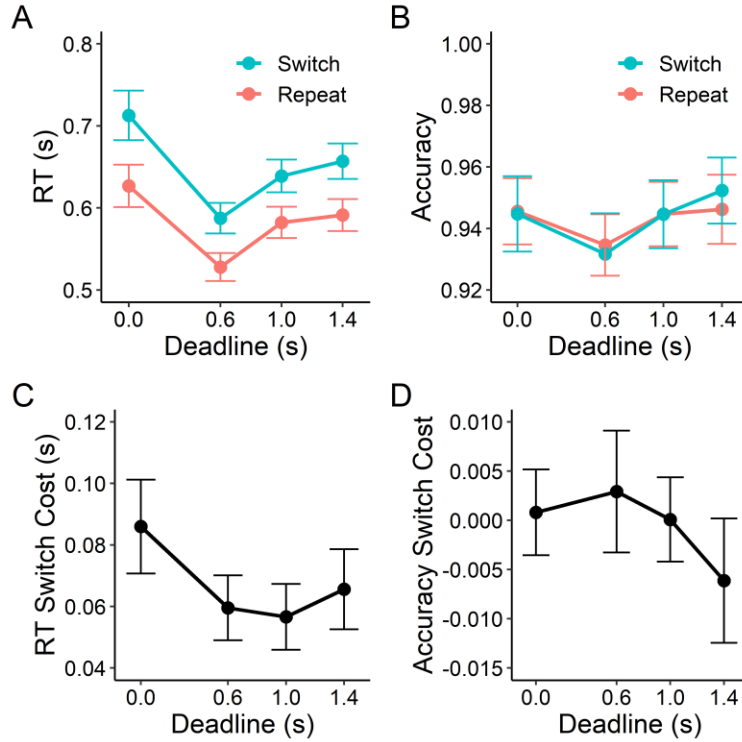


Figure 6: Experiment results. (A) Mean RT and (B) accuracy for switch and repeat trials as a function of deadline. (C-D) Corresponding RT and accuracy switch costs (switch minus repeat for RT, repeat minus switch for accuracy). Error bars indicate ± 1 standard error of the mean.

4.1.3 Model Fitting

To quantify the relative effects of task preparation and speed-accuracy trade-off, we fit four versions of our model to the empirical data, that differed in which parameters were allowed to vary over deadline conditions. The parameters (and the effects they were meant to capture) were: number of CSI time steps (duration of preparation), gain of the control unit (intensity of preparation), DDM threshold (speed-accuracy trade-off), and non-decision time (basic sensory encoding and motor execution). In the first version, all four parameters were allowed to vary for a given participant across deadline conditions. In the second version, preparation duration was fixed for a given participant (based on their average estimate over all deadline conditions from the previous model fit) while gain, threshold, and non-decision time were allowed to vary. In the third version, both preparation duration and gain of the control units were fixed (based on previous estimates), while threshold and non-decision time were allowed to vary. Finally, in the fourth version preparation duration, gain, and threshold were fixed across conditions and only non-

decision time was allowed to vary. We then applied Bayesian model selection to the fits of these four model versions, to determine which provided the best fit to the data while discounting for the number of free parameters. Because the model is intractable to closed form analysis, we used an approximated maximum likelihood estimation (MLE) approach to fit the model non-hierarchically to empirical data. We refer the reader to the Appendix A.3 for additional details regarding the fitting procedure, an overview of and rationale behind other fixed parameter values, and parameter recovery tests.

Parameter Estimates. We first report the fit of the first version of the model, in which all of the following parameters — preparation duration, control unit gain, and DDM threshold — as well as non-decision time were allowed to vary (see Fig. 7 for an illustration of fitted parameters from this first version of the model). Applying repeated measures ANOVA to the fitted parameter values with deadline as within-subjects factor indicated a significant effect on threshold ($F(1.81, 54.23) = 19.33, p < 0.001, \eta_p^2 = 0.392$; see Fig. 7A). Pairwise comparisons revealed significant differences between all deadlines except 1.0 s and 1.4 s, ($p = 0.292$, all other $ps \leq 0.001$), consistent with an adjustment of the speed-accuracy trade-off to the deadline in the predicted U-shaped manner. There was also a significant effect on preparation duration ($F(3, 90) = 3.10, p = 0.221, \eta_p^2 = 0.094$; see Fig. 7B), although pairwise comparisons indicated significant or marginally significant differences between only the 0 and the other deadlines (0.6, 1.0, and 1.4 s deadlines, $ps = 0.017, 0.060, 0.020$, respectively), which did not differ significantly from one another ($ps \geq 0.529$). Similarly, there was a significant effect on control unit gain ($F(3, 90) = 4.36, p = 0.006, \eta_p^2 = 0.127$; see Fig. 7C). Again, the only significant difference in pairwise comparisons was between 0 and the other deadlines (0.6, 1.0, and 1.4 s, $ps = 0.003, 0.013, 0.003$, respectively), with all other comparisons $p \geq 0.497$. Lastly, there was a significant effect on non-decision time ($F(2.18, 65.27) = 13.18, p < 0.001, \eta_p^2 = 0.305$; see Fig. 7D). Pairwise comparisons revealed significant differences between all deadlines ($ps \leq 0.010$) except 1.0s and 1.4s ($p = 0.845$).

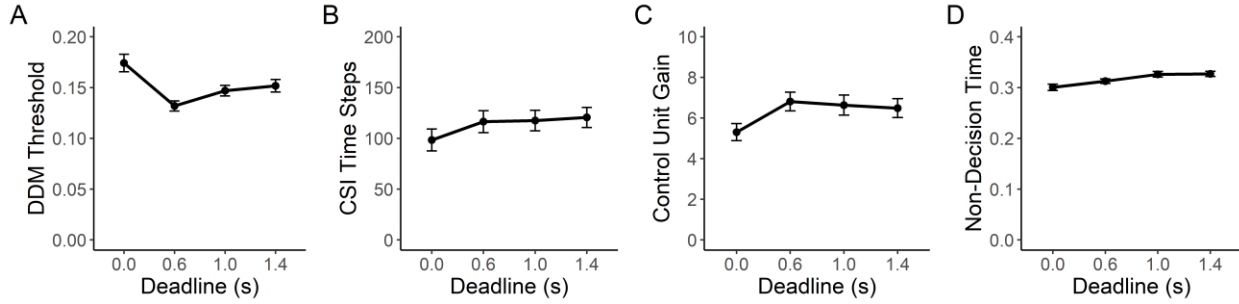


Figure 7: Fitted parameter values of the first version of the model, where each was a free parameter. (A) DDM threshold, (B) CSI time steps, (C) gain of control units, and (D) non-decision time as a function of deadline. The illustrated effects of deadline on parameter estimates did not qualitatively change when we fixed the parameters in a stepwise manner (see text for more detail) and thus we only show here the estimates from the first version of the model. Error bars indicate ± 1 standard error of the mean.

Next, we repeated the model fits while fixing, in a stepwise manner, preparation duration, control unit gain, and DDM threshold across deadline conditions. We then used Matlab's Statistical Parametric Mapping package to apply Bayesian model selection (Rigoux et al., 2014; Stephan et al., 2009) to BIC scores of the four versions of the model. The Bayes Omnibus Risk was 0.15 and the model with fixed preparation duration and control gain but varying threshold and non-decision time was 10 times more probable than the other models. This indicates that a model with deadline-insensitive task preparation (both duration and intensity), coupled with strategic adjustment of the speed-accuracy trade-off, provides the best account of the data.

Table 1. Bayesian model comparison results.

Fixed parameters across deadlines	Protected exceedance probability	Bayes omnibus risk
None	0.04	0.15
CSI time steps	0.07	
CSI time steps & gain	0.81	
CSI time steps & gain & threshold	0.08	

Highest probability model is bolded.

4.1.4 Experiment Discussion

Our experiment showed that, with a parametric, within-subject manipulation of response deadline for earning a monetary reward, RT (and switch costs) follow the general predictions made by our model. Specifically, each dependent measure exhibited a U-shaped curve as response deadline increased, with RTs additionally exhibiting a predicted, more pronounced U-shape for switch than in repeat trials. Importantly, the observation of similar U-shaped curves in RTs and accuracy suggest the presence of a speed-accuracy trade-off.

Model selection favored a version of the model in which task preparation was fixed across deadline conditions, but speed-accuracy trade-off varied. These findings highlight that an interaction between speed-accuracy trade-off and task switches versus repeats can qualitatively create the impression of strategic adjustment of task preparation in RT, that is not supported by quantitative fits using a dynamical systems model of task switching performance.

5 General Discussion

We developed a computational model of task switching experiments meant to capture relevant constraints and choices intrinsic to such experiments, including when to prepare for an upcoming task and how to balance the speed-accuracy trade-off when confronted with a response deadline. We demonstrated the overall plausibility of this model in terms of its ability to reproduce patterns in existing data; we also demonstrated that relying on the mixture property of RT distributions (De Jong, 2000; Nieuwenhuis & Monsell, 2002) provides inconclusive evidence for a role of task preparation versus strategic adjustments of the speed-accuracy tradeoff when participants must respond by a deadline. Specifically, we showed that both of these factors can influence switch costs, raising questions about the extent to which these can be used to infer preparation without considering decision threshold (i.e., the speed-accuracy tradeoff). To demonstrate this, and provide further insight into the strategic adjustments individuals make in task switching experiments, we conducted an experiment that employed a within-subject design and parametrically manipulated the response deadline in a task switching paradigm. This experiment replicated previous findings that overall task performance, and especially switch costs, change across deadline conditions. We then used model comparison to show that these changes were best explained by adjustments in the speed-accuracy tradeoff, and *not* preparation duration

or intensity. This provides an alternative to the interpretation of previous studies suggesting that switch costs — and the residual switch cost in particular — reflect strategic constraints on task preparation, instead suggesting that these may reflect strategic changes in managing the speed-accuracy tradeoff. In the remainder of this discussion, we consider how these findings, and our model, relate to and may further inform the role of strategic control over task preparation.

5.1 Motivational Effects in Task Switching

In the Introduction, we highlighted a general distinction between structural interpretations of the residual switch cost, which emphasize the role of constraints imposed by the cognitive architecture, and strategic interpretations which instead emphasize choices based on costs and benefits. Correspondingly, these different interpretations differ with respect to whether task preparation is expected to be sensitive to experimental manipulations. The model that we developed here does not fall cleanly into either class of interpretation. The model is “structural” in the sense that switching between tasks requires a nontrivial amount of time due to attractor dynamics in task preparation. However, the model is “strategic” in the sense its task preparation and decision parameters also can be allowed to vary across experiment conditions to reflect sensitivity to strategic control. Nevertheless, the model provides good fits to empirical data when its task preparation parameters are fixed across experiment conditions, suggesting these are indeed structural (or at least not adjusted strategically).

More generally then, it is necessary to distinguish between at least two obvious forms of strategic control in task switching: effort minimization and the speed-accuracy trade-off. The impact of effort minimization strategies on behavior has previously been considered in a wide range of literature (e.g., (Ballard et al., 1997; Bettman et al., 1990; Botvinick et al., 2009; Botvinick & Braver, 2015; Droll & Hayhoe, 2007; Johnson & Payne, 1985; Kool et al., 2010; McGuire & Botvinick, 2010; Shah & Oppenheimer, 2008; Shenhav et al., 2017; Wilcox, 1993, 1994)). In the task switching literature, De Jong (2000) and Nieuwenhuis and Monsell (2002) posited that residual switch costs are driven by mental effort associated with preparing for a task switch. However, the experimental and model-fitting results in the current work failed to provide evidence for strategic adjustments in the *timing* of the preparation process, since results were more consistent with fixed CSI time steps — albeit coupled to a decision process in which the speed-accuracy trade-off could be adjusted.

Aside from timing of the preparation process, another potential source of effort minimization effects in task switching is the *strength* of preparation, as has been posited by Yeung and Monsell (2003). They argued that task preparation is graded, with stronger preparation requiring more effort but translating into a higher response process drift rate — and thus to responses that are both faster and more accurate. To model the intensity of task preparation, we drew upon the work and formal analysis of Musslick et al. (2018, 2019). They formalized the strength of task preparation in terms of constraint on control intensity within a dynamical system, and they demonstrated that this can be used to regulate the trade-off between cognitive stability and flexibility (Cools, 2019; Goschke, 2003; Hommel, 2015). Increasing control intensity increases the maximal strength of task preparation, thereby maximizing performance on the prepared task and minimizing distraction. However, this comes at the cost of slower task reconfiguration when the environment changes. Musslick et al. (2018, 2019) showed that optimal intensity of task preparation is lower (i.e., limiting the strength of task preparation) as the demand for flexibility increases, and found that participants indeed adopted lower intensity of control when the probability of task switches was high as compared to low. However, participants showed overall lower intensity of control than was predicted to be optimal, which could be indicative of the effort minimization proposed by Yeung and Monsell 2003). This leaves open the possibility that such effort minimization might be offset by incentives like monetary reward.

We tested whether participants adjust their intensity of task preparation when doing so increases their chance of beating a deadline and thereby earn a reward. We updated the dynamical system model of Musslick et al. (2018, 2019) and integrated it within the network architecture of Cohen et al. (1990), who showed that their model is suitable to explain many phenomena associated with attention and controlled stimulus processing (for details on how our implementation differs from these previous models, see the Appendix A.2). Assuming effort-sensitive, strategic constraints on task preparation, one might expect that, when incentivized to meet a stringent but barely manageable response deadline, participants would increase strength of task preparation and perhaps also initiate their preparation earlier during the CSI to compensate for the resulting slower switching between tasks. However, model fitting and model selection suggested the data was best accounted for by adjustments of the speed-accuracy tradeoff (i.e., decision threshold) but not the strength (i.e., control intensity) nor the duration of preparation.

Nevertheless, other work has hinted at an effort-performance trade-off by showing that

monetary incentives during task switching can increase speed without leading to a decrease in accuracy (Kleinsorge & Rinkenauer, 2012; Savine et al., 2010). Given the current results, an important direction for future work is to understand which types of experimental manipulations elicit a standard speed-accuracy trade-off, as we observed in the present experimental data and model fits, and which types of manipulations “break” this trade-off (cf. (Manohar et al., 2015), and directly impact the decision to prepare for a switch trial.

As already discussed in Section 4, one factor may be the size of monetary reward. In our own experiment, which used rewards on the order of \$10/hour, we were able to reduce but not eliminate the residual switch cost. Importantly, our model simulations do predict that if participants can be incentivized to maximize task preparation during a long CSI then this should be able to obliterate entirely the residual switch cost. Interestingly, Savine et al. (2010) reportedly achieved this by employing a categorical monetary incentive without actually specifying to participants the incentive amount, which may have led participants to make optimistic assumptions about actual payoff magnitudes and thus motivate them more strongly than in our own experiment. While Kleinsorge and Rinkenauer’s Experiment 1 (2012) found that RT switch costs were eliminated with a magnitude of monetary incentive that was both effectively signaled to participants and rather low (on average approximately \$2, in addition to a fixed amount of about \$12, for a session of approximately 1 hour), they might have been statistically underpowered with only 8 participants per group. Thus, an important direction for future work would be to determine whether parametric manipulation of incentive can monotonically decrease and eliminate the residual switch cost.

Another possibility is that *extrinsic* motivators such as monetary reward are less suited to probe the strategic origin of the residual switch cost than other, *intrinsic* motivators. As recently proposed by Blain and Sharot (2021), intrinsic reward does *not* involve a primary rewarding stimulus (e.g., food and water) or a secondary rewarding stimulus (e.g., money) and therefore does not seem to provide a direct benefit to survival. Instead, an intrinsically rewarding activity (e.g., playing, solving crossword puzzles) is one that is performed in its own right without necessarily leading to an external outcome. An activity might be experienced as intrinsically rewarding when it confers an increased sense of progress and self-efficacy (Blain & Sharot, 2021). Consistent with this notion, it was recently observed that people are willing to engage in more frequent task-switching to obtain information on their progress through the experiment (Devine & Otto, 2022). These findings suggest that the experience of progress and increased self-efficacy are relevant

motivators of cognitive effort and future work should explore whether the experience of intrinsic reward can reduce or eliminate the residual switch cost without sacrificing response accuracy.

Lastly, it remains possible that the residual switch cost is due entirely to structural constraints in task preparation that are insensitive to strategic control. One novel explanation that could account for the residual switch cost purely in structural terms is based on the experience of ‘task conflict’. That is, when presented with a cue for a task-switch, the cognitive system must reconfigure its task-sets and during this reconfiguration process there is a period of time where multiple task-sets are at least moderately active. This can be seen in Figure 2D where we illustrate the dynamics of task-set reconfiguration: halfway through the task reconfiguration process, both control units of the model are equally active. Such co-activation of task-sets could be detected by the brain as task conflict, c.f. (Kalanthoff et al., 2018), and elicit an adaptive increase in response threshold (Cavanagh & Frank, 2014) to buy time and ensure the response process does not complete before the task conflict has resolved. In other words, it is possible that the residual switch cost might not indicate a limitation of cognitive flexibility but instead reflect an adaptive response to task conflict resulting from task-set inertia (cf. (Allport et al., 1994). This hypothesis is supported by the observation that threshold increases on task-switch trials, even when CSI is long (Schmitz & Voss, 2012, 2014). As we showed earlier in Section 3, increasing the response threshold increases the switch cost in RT and thus might give rise to a residual switch cost arising from response caution. This would furthermore be consistent with the observation that residual switch costs do not occur in accuracy, as switch and repeat accuracy converge at higher thresholds (Fig. 3N). Future work should examine if a switch-induced increase in threshold can indeed be tied to the presence of task-conflict.

5.3 Conclusion

We presented a model of the classical task switching paradigm in terms of the constraints and choices faced by participants and used this model to simulate scenarios that differed with respect to strategic control over the task preparation process, instantiating examples that aligned with structural and/or strategic explanations of the residual switch cost. Applying model fits to new empirical data favored a model where initiation and strength of the preparatory process was fixed across conditions, albeit coupled with strategic adjustment of the speed-accuracy trade-off. Although one might interpret this as supporting structural interpretations of the residual switch

cost, we caution against strong conclusions one way or the other. We have specifically argued that reducing the residual switch cost using small monetary reward, contingent on beating a deadline, does not provide evidence for strategic constraints on task preparation. However, we leave open the possibility that larger incentives, or other types of manipulations, might expose strategic constraints on task preparation. More generally, we hope that the model presented in this paper helps illustrate the value of detailed cognitive modeling and motivates a continued effort to develop formally explicit models of performance in the task switching paradigm and more precisely characterize the mechanisms responsible for cognitive control.

Appendix

In this appendix we begin by illustrating evidence for a speed-accuracy trade-off in early work on monetary reward and response deadlines in task switching. Then we describe how, and motivate why, our computational model of task switching is different from the two models upon which it is based. This is followed by an overview and discussion of free and fixed parameters of the model, and a brief description of the approximated likelihood estimation method for model fitting. Lastly, we report parameter recovery tests and compare empirical behavior to simulated behavior based on model fits.

A.1 Speed-Accuracy Trade-Off in Monetary Incentives

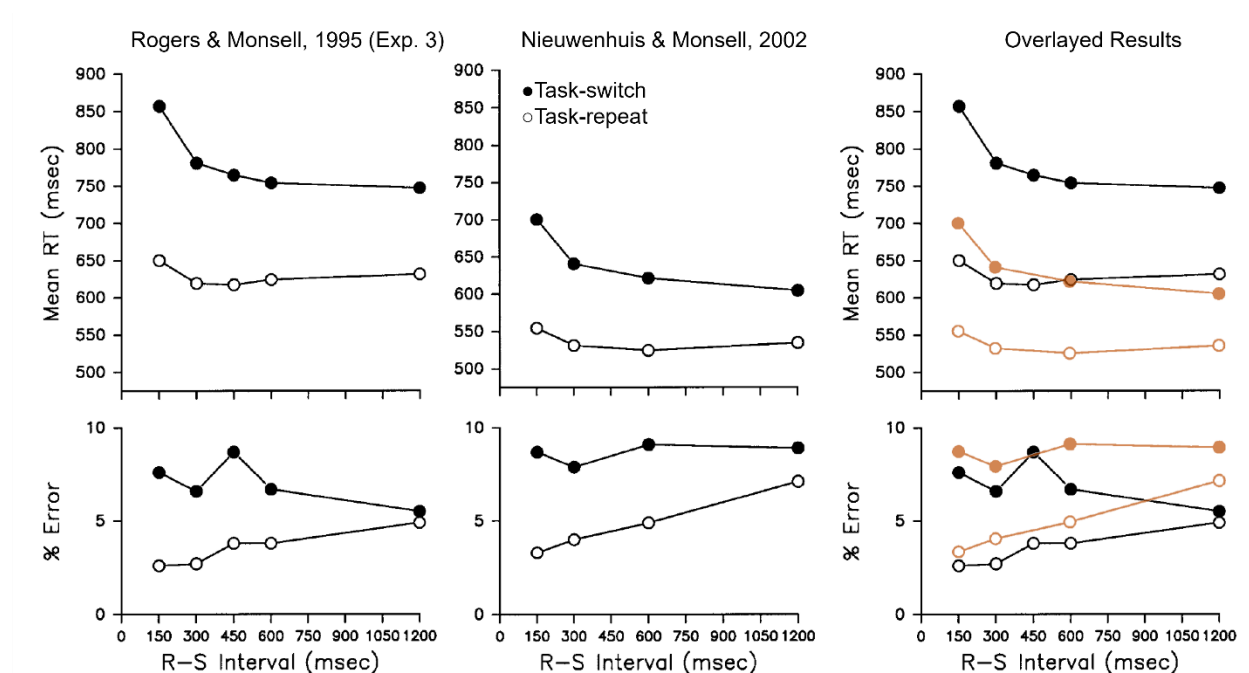


Figure S1: Behavioral results as reported in Nieuwenhuis and Monsell (2002), plus an overlay of the first two panels, with Nieuwenhuis and Monsell's incentivized experiment results in orange. For precision, we show the exact figures as shown in Nieuwenhuis and Monsell (2002). RTs are lower and error rates are higher in the incentivized experiment, indicating that the combined manipulation of reward and response deadline might have induced an emphasis on speed at the cost of accuracy.

A.2 Relation to Previous Models

Here we explain how and why our model implementation differs from its foundational building blocks: the dynamical system model of task preparation by Musslick et al. (2018, 2019) and the network model of controlled stimulus processing by Cohen et al. (1990).

First, we describe the differences between the control units of Musslick et al. versus the present implementation. A quantitative difference is that in the model of Musslick et al., control units update their activation once per trial in a single large time step, whereas the present units update their activation multiple times per trial across several smaller time steps. This allows us to model the activation dynamics at a finer, within-trial timescale. Additionally, in the model of Musslick et al., the preparation and response processes were coupled in a serial manner, i.e., on each trial the control units updated their activation once and then their activation was translated into a constant drift rate for that trial. In our model, control units continue to update after the DDM process starts, which allows us to capture the effect of ongoing task preparation during the response process through a time-varying drift (cf. (Liu et al., 2008; Shvartsman et al., 2015; Srivastava et al., 2017)).

A qualitative difference in the activation dynamics stems from self-recurrent excitation and inhibition in the control units of Musslick et al. As a result, upon a task switch, the rise in activity for the new task is *faster* than the drop in activity for the old task, which produces an upward bow in the activity trajectory relative to the diagonal of the state space (Fig. S2A). In contrast, we implement net leakage and inhibition in our units². As a result, upon a task switch the rise in activity for the new task is *slower* than the drop in activity for the old task, which produces a downward bow in the activity trajectory (Fig. S2B). This distinction is crucial because these trajectories produce qualitatively different interactions between task switching and the response-congruence of a stimulus.

² In the LCA, self-excitation and decay function together and so can be reduced to a single parameter that leads to net leakage (when self-excitation < decay) or self-amplification (when self-excitation > decay).

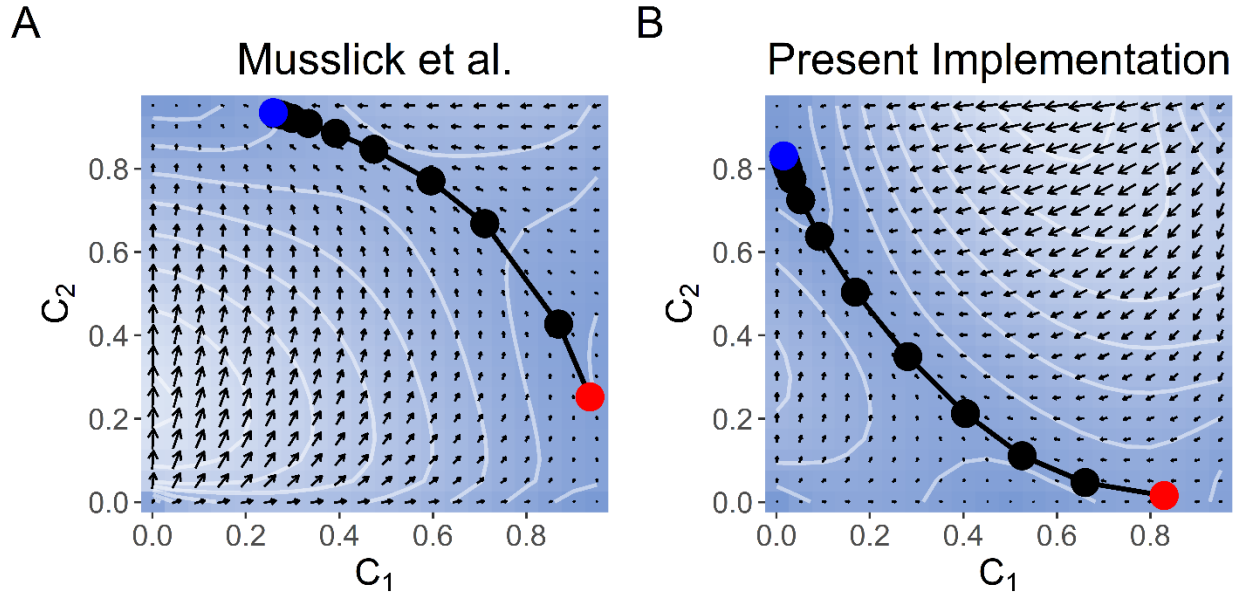


Figure S2: Control activity trajectory. Phase portraits for activity of the control units in in Musslick et al. (2018, 2019) (A) and the present model (B) in response to task cue inputs, showing trajectory from prior activation (e.g., red dot for task 1) to the cued task (e.g. blue for task 2). Contour lines and arrows indicate the energy and shape of the attractor landscape after a task switch from task 1 to task 2. Note that in (A) the trajectory has an upward curve relative to the diagonal, whereas in (B) the trajectory has a downward curve.

Note that in the case of a congruent stimulus, the activities of both control units bias response selection towards the same response. Consequently, the upward trajectory results in stronger biasing for the correct response on congruent task *switches* than congruent task *repeats* (i.e., the sum of control activities is larger upon a task switch than repeat). This produces a negative (reversed) switch cost on congruent trials. The downward trajectory avoids this issue and instead produces smaller but still positive switch costs on congruent trials, due to weaker biasing of the correct response on congruent task switches than congruent task repeats (i.e., the sum of control activities is now smaller during a task switch than repeat). Only this downward trajectory reproduces the interaction between task switching and incongruence effects established in the literature (Goschke, 2000; Rogers & Monsell, 1995) and observed, albeit marginally significant, in our own data.

Another qualitative difference from Musslick et al. relates to congruence effects as well. In

their model implementation, the drift rate of a DDM is a linear combination (i.e., sum) of an automatic and controlled component

$$drift = \omega_a(S_1 + S_2) + (S_1C_1 + S_2C_2) \quad (6)$$

The automatic component $\omega_a(S_1 + S_2)$ is simply a weighted sum of two stimulus inputs of equal strength $S \in \{1, -1\}$. On congruent trials, these stimulus inputs have the same sign and thus sum to a nonzero value, whereas on incongruent trials the stimulus inputs have opposite sign and thus sum to 0. The controlled component $S_1C_1 + S_2C_2$ is similar but here stimulus inputs are weighed by their corresponding control unit activity. Consequently, drift rates are large on congruent trials where the identical sign of stimulus inputs leads to summation of values within the automatic and controlled component that is always in the same numerical direction. In contrast, drift rates are much smaller on incongruent trials because the opposite sign of stimulus inputs cancels out the automatic component while the controlled component is now the difference of control unit activities. In early iterations of our model simulations and model fits we observed that this pronounced difference in drift on congruent versus incongruent trials led the model to produce incongruence costs that were much larger than observed in empirical data. This suggested that nonlinear rather than linear combination of stimulus inputs and control unit activity might better capture empirical behavior.

For this reason, we integrated our control units within the network architecture of Cohen et al. (1990). Their model was designed precisely to capture nonlinearities in automatic and controlled stimulus processing. They framed their model in terms of the classical Stroop task and thus it includes two processing pathways that respectively map word color and word meaning onto shared response units. In each pathway, binary stimulus inputs map onto intermediary hidden units, which in turn map onto response units. Both hidden and response units have a nonlinear, sigmoid activation function and the hidden units include a negative bias term in their input. This negative bias limits the unit's sensitivity to stimulus input unless a corresponding control unit is activated and offsets the bias. This effectively "turns on" the unit by driving its activation towards the steep, middle part of the sigmoid function's slope where it is maximally responsive to input.

The nonlinearity in this model is crucial as it introduces a ceiling on the activation of the correct response unit and imposes an asymmetry between the facilitation of response selection on congruent trials and the interference on incongruent trials: the response units already reach the non-linear part of their activation function when receiving excitatory input from task-relevant

hidden units, thus any additional excitatory unput from the irrelevant hidden units only modestly increases response activation. For a more thorough consideration of this point we refer the reader to Cohen et al. (1990) and limit ourselves here to emphasizing that the asymmetry between facilitation and interference reduces the difference in drift on congruent and incongruent trials as compared to Musslick et al.'s model implementation. This improves quantitative fit to incongruence costs observed in empirical data.

In summary, we adopted the model by Cohen et al. (1990) with the following differences: we implement balanced connection weights in the two processing pathways for digit parity and magnitude whereas they implemented unbalanced weights to capture the differential strength of association between word meaning and word color with their corresponding response; in our model the difference in activation of response units serves as the drift rate in a DDM; in our model the control units are LCAs instead of binary control units. This results in the dynamical system model of task switching as we described previously.

To demonstrate the added value of LCA control units over Cohen et al.'s simpler, binary control units, in terms of accounting for empirical data, we compared the dynamical system model fits from Section 5 in the main text (where preparation time steps and gain were fixed parameters across deadline conditions) with new model fits where the control units were binary as in the case of Cohen et al. Bayesian model selection suggested very strong support for the dynamical system implementation, as the Bayes Omnibus Risk was 0.011 and protected exceedance probabilities for the dynamical system model versus the binary control model were 0.994 versus 0.006, respectively.

A.3 Model Fitting

Free & Fixed Parameters. For model fits reported in the main text, free parameters and their respective lower and upper bounds included: DDM threshold (0.01, 0.5), CSI time steps (0, 200), gain of control units (1, 10), and non-decision time (0.1, 0.4). Fixed parameter values for control units were decay = 3, inhibition = 3, and time step size = 0.01; for DDM were time step size = 0.001 and noise = 0.05; connection weight from stimulus input to hidden units = 1.3 and hidden unit inputs to response units = 2. We note that this combination of fixed values is probably not unique in accounting for our data, but below we provide a thorough rationale for how we

arrived at these values.

It is necessary to fix values for decay and inhibition because they trade off against each other at least in some parts of the parameter space of a two-unit LCA, which significantly complicates parameter fitting (Miletić et al., 2017). However, we benefit from a unique advantage over previous attempts at fitting the LCA: we can constrain our choice of parameter values based on the activity trajectory of control units, as discussed in A.2. Specifically, the trajectory requires a downward curve to reproduce the desired interaction between task switching and response-congruence. If the downward curve is too strong then the difference in switch costs on congruent and incongruent trials becomes too large, but if the downward curve is too weak it approaches a straight line³ and eliminates differences in switch costs. Both cases violate the pattern observed in our data and existing literature (Goschke, 2000; Rogers & Monsell, 1995). Nevertheless, larger values of decay or inhibition can also be absorbed by rescaling other model parameters. Therefore, as a rule of thumb we chose the lowest values for decay and inhibition that give a downward curve in the activity trajectory and produce congruence effects as observed in empirical data. This involved several iterations of model simulation and model fitting.

To summarize this process, we observed that inhibition must be larger than 1 to produce a downward curve in the activity trajectory. For simplicity we constrained decay and inhibition to be equal. When decay is set at 2 or lower there is a very slow drop in activity for the old task upon a task switch, resulting in a strong downward curve that overproduced congruence effects. Conversely, decay of 4 or higher resulted in a trajectory that increasingly approached a straight line and thus underproduced congruence effects. Therefore, we set decay and inhibition to 3, which resulted in the modest downward curve as shown in Figure 2D of the main text and Figure S2B in the previous section.

It is then necessary for control unit time step size to be larger than DDM time step size, because if control unit time step size is too small, the units are slow in updating their activation and switch errors are overproduced. Conversely, if the control unit time step size is too large then task preparation saturates so quickly it produces negligible switch costs. Therefore we set DDM time step size to 0.001 and through several iterations of model simulation and fitting settled on control unit time step size of 0.01. A DDM noise of 0.05 allowed the model to reproduce the high

³ A noteworthy example of the trajectory becoming a straight line is the special case where decay and inhibition of a two-unit LCA are both equal and sufficiently high. The LCA then reduces to a DDM (Bogacz et al., 2006).

but not perfect levels of accuracy (about 90% or greater) typically observed in empirical data.

Lastly, we set the connection weight from hidden units to response units at 2, roughly similar to Cohen et al. (1990) for their non-automatized (i.e., word color) task. This gives the aforementioned asymmetry in facilitation on congruent trials versus interference on incongruent trials. We then fit a separate pilot data set of parity-magnitude task-switching to fit the connection weight from stimulus inputs to hidden units, which was found to be on average 1.3 and thus, consistent with the findings of Cohen et al. (1990), limits the extent to which stimulus inputs can activate hidden units in the absence of excitatory input from the control units.

Approximated Likelihood Estimation. Model fitting is complicated because the LCA is intractable and in our model coupled to a time-varying DDM drift rate. Therefore we relied on an approximated maximum likelihood estimation method proposed by Turner and Sederberg (2014). In principle, this involves 1) simulating the model many times for a proposed set of parameter values, 2) applying probability density estimation to the simulation results to approximate the likelihood function, and 3) using an optimization algorithm to minimize the negative log-likelihood. The more often the model is simulated for a set of parameter values, the smoother and more reliable the approximated likelihood function becomes. We therefore simulated the model 40,000 times per proposed set of parameters and used GPU-optimized, custom-written code to achieve feasible duration of the model fitting process. To minimize the log-likelihood we used the differential evolution function from the Python library SciPy.

The already-large number of simulations required to fit a single participant’s data quickly grows even larger when fitting more trials and/or parameters, which prevented hierarchical data fitting (although we do not rule out the possibility that future advances in code optimization might make hierarchical fitting a realistic endeavor). Therefore, we fit the data from our deadline experiment separately per participant and deadline condition. To still allow a parameter to be fit per participant but be fixed across deadline conditions (e.g., the number of preparation time steps), we first fit the parameters separately across conditions and then fixed, in a step-wise manner, each parameter based on its averaged estimate across conditions obtained in earlier model fits. When comparing BIC scores for model selection we summed the BIC scores across conditions for each participant.

Parameter Recovery. To determine parameter recoverability we simulated 100 datasets

using the free parameter ranges and fixed parameter values as described at the start of this section⁴. Each dataset included 384 trials, similar to the number of trials per deadline condition in our experiment. To quantify recoverability we regressed the simulated (‘true’) parameter values against the fitted (‘recovered’) values using both standard and robust linear regression (Fig. S3).

Recovery is excellent for each parameter as indicated by (unstandardized) beta coefficients that are close to 1.0: DDM threshold $b_{standard} = 0.96, b_{robust} = 0.96$, CSI time steps $b_{standard} = 0.67, b_{robust} = 0.76$, gain $b_{standard} = 0.97, b_{robust} = 0.99$, and non-decision time $b_{standard} = 1.00, b_{robust} = 1.00$, all $ps < 0.001$. Visual inspection suggests that recovery of CSI time steps is noisier than the other parameters, but the high robust regression coefficient suggests that recovery is nevertheless good for the majority of cases. The residuals of recovered CSI time steps did not correlate with the other parameter residuals, all $ps > 0.118$, suggesting no systematic trade-offs in the recovery of CSI time steps. Visual inspection also suggests that recovery is noisier for higher levels of gain, which can be explained by the sigmoid activation function of control units getting increasingly squashed against its absolute minimum of 0 and maximum of 1. Thus the activation values become very similar for high levels of gain. The variance in recovery of gain is unbiased and averages out across the simulated datasets, which is why the regression coefficients remain close to 1. Note that, as shown in the main text, average fitted gain values from our empirical data are in the 5 to 7 range where recovery is good.

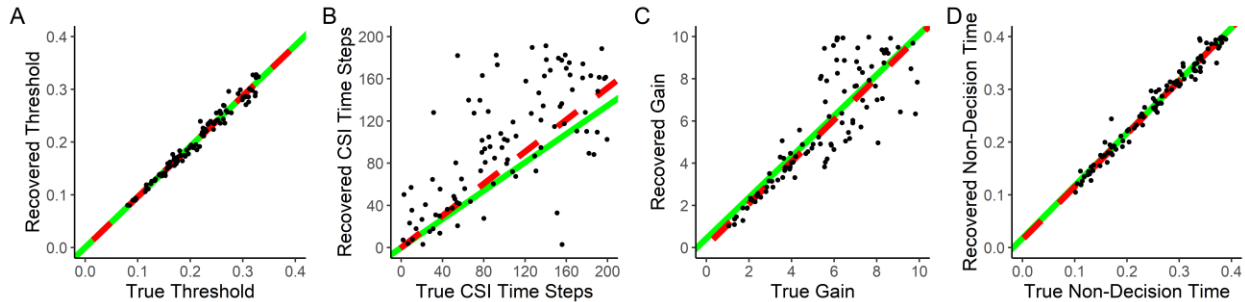


Figure S3: Parameter recovery of (A) DDM threshold, (B) CSI time steps, (C) gain of control units, and (D) non-decision time. Solid green lines illustrate standard linear regression, dashed red lines illustrate robust linear regression.

⁴ We used a somewhat smaller range in thresholds of 0.08 to 0.33, based on the range in threshold estimates from empirical data. This was done because the other parameters became less recoverable at very low or very high thresholds, since in both cases the more nuanced impact of task preparation parameters become less apparent.

Empirical versus model-simulated behavior. To determine whether the model reproduces key patterns observed in our empirical data, we used the model-fitted parameter estimates to simulate behavior and plot it alongside the empirical data. The model captures the U-shaped response to deadline in terms of RT (Fig. S4A, B) and accuracy (Fig. S4C, D). Model-simulated switch costs are smaller in RT (Fig. S4E, F) and larger in accuracy (Fig. S4G, H), but both reproduce the U-shaped response from the 0 to 1.0 s deadline. Thus, the model captures the smaller switch costs in RT but larger in accuracy in the intermediate response deadline condition.

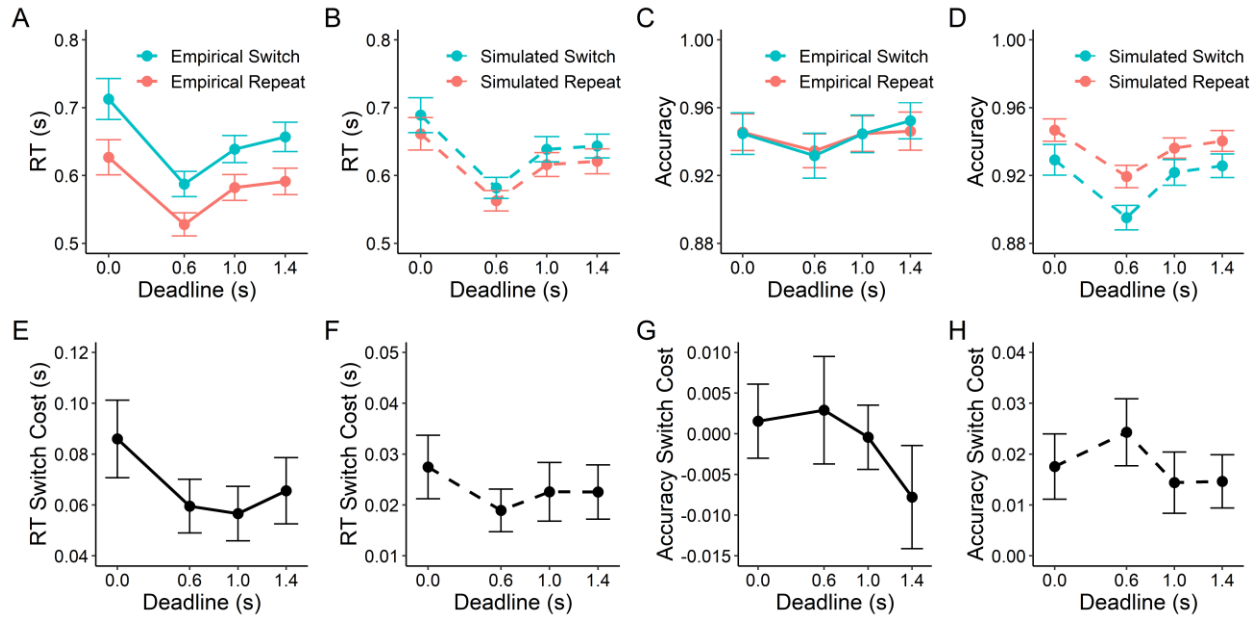


Figure S4: Empirical data alongside model-simulated data. (A,B) Mean RT and (C,D) accuracy for switch and repeat trials as a function of deadline. (E-H) Corresponding RT and accuracy switch costs (switch minus repeat for RT, repeat minus switch for accuracy). Solid lines are empirical data, dashed lines are simulated data. Error bars indicate ± 1 standard error of the mean.

References

- Allport, D. A., Styles, E. A., & Hsieh, S. (1994). Shifting intentional set: Exploring the dynamic control of tasks. In *Attention and performance 15: Conscious and nonconscious information processing* (pp. 421–452). The MIT Press.
- Altmann, E. M., & Gray, W. D. (2008). An integrated model of cognitive control in task switching. *Psychological Review*, 115(3), 602–639. <https://doi.org/10.1037/0033-295X.115.3.602>
- Aston-Jones, G., & Cohen, J. D. (2005). An integrative theory of locus coeruleus-norepinephrine function: Adaptive gain and optimal performance. *Annual Review of Neuroscience*, 28, 403–450. <https://doi.org/10.1146/annurev.neuro.28.061604.135709>
- Balci, F., Simen, P., Niyogi, R., Saxe, A., Hughes, J. A., Holmes, P., & Cohen, J. D. (2011). Acquisition of decision making criteria: Reward rate ultimately beats accuracy. *Attention, Perception, & Psychophysics*, 73, 640–657. <https://doi.org/10.3758/s13414-010-0049-7>
- Ballard, D. H., Hayhoe, M. M., Pook, P. K., & Rao, R. P. (1997). Deictic codes for the embodiment of cognition. *The Behavioral and Brain Sciences*, 20(4), 723–742; discussion 743–767. <https://doi.org/10.1017/s0140525x97001611>
- Bettman, J. R., Johnson, E. J., & Payne, J. W. (1990). A componential analysis of cognitive effort in choice. *Organizational Behavior and Human Decision Processes*, 45(1), 111–139. [https://doi.org/10.1016/0749-5978\(90\)90007-V](https://doi.org/10.1016/0749-5978(90)90007-V)
- Blain, B., & Sharot, T. (2021). Intrinsic reward: Potential cognitive and neural mechanisms. *Current Opinion in Behavioral Sciences*, 39, 113–118.

<https://doi.org/10.1016/j.cobeha.2021.03.008>

Bogacz, R., Brown, E., Moehlis, J., Holmes, P., & Cohen, J. D. (2006). The physics of optimal decision making: A formal analysis of models of performance in two-alternative forced-choice tasks. *Psychological Review*, 113(4), 700–765. <https://doi.org/10.1037/0033-295X.113.4.700>

Botvinick, M. M., & Braver, T. (2015). Motivation and cognitive control: From behavior to neural mechanism. *Annual Review of Psychology*, 66, 83–113.
<https://doi.org/10.1146/annurev-psych-010814-015044>

Botvinick, M. M., Huffstetler, S., & McGuire, J. T. (2009). Effort discounting in human nucleus accumbens. *Cognitive, Affective & Behavioral Neuroscience*, 9(1), 16–27.
<https://doi.org/10.3758/CABN.9.1.16>

Brainard, D. H. (1997). The Psychophysics Toolbox. *Spatial Vision*, 10(4), 433–436.

Brown, S., Lehmann, C., & Poboka, D. (2006). A critical test of the failure-to-engage theory of task switching. *Psychonomic Bulletin & Review*, 13(1), 152–159.
<https://doi.org/10.3758/BF03193827>

Cavanagh, J. F., & Frank, M. J. (2014). Frontal theta as a mechanism for cognitive control. *Trends in Cognitive Sciences*, 18(8), 414–421. <https://doi.org/10.1016/j.tics.2014.04.012>

Cohen, J. D., Dunbar, K., & McClelland, J. L. (1990). On the control of automatic processes: A parallel distributed processing account of the Stroop effect. *Psychological Review*, 97(3), 332–361. <https://doi.org/10.1037/0033-295x.97.3.332>

Cools, R. (2019). Chemistry of the Adaptive Mind: Lessons from Dopamine. *Neuron*, 104(1), 113–131. <https://doi.org/10.1016/j.neuron.2019.09.035>

- da Silva Castanheira, K., LoParco, S., & Otto, A. R. (2021). Task-evoked pupillary responses track effort exertion: Evidence from task-switching. *Cognitive, Affective, & Behavioral Neuroscience*, 21(3), 592–606. <https://doi.org/10.3758/s13415-020-00843-z>
- De Jong, R. (2000). An intention-activation account of residual switch costs. In S. Monsell & J. Driver (Eds.), *Attention and performance xviii: Control and cognitive processes* (pp. 357–376). MIT Press. [https://research.rug.nl/en/publications/an-intentionactivation-account-of-residual-switch-costs\(bc0f7750-f8ff-469c-8111-0a3836887d38\).html](https://research.rug.nl/en/publications/an-intentionactivation-account-of-residual-switch-costs(bc0f7750-f8ff-469c-8111-0a3836887d38).html)
- De Jong, R., Berendsen, E., & Cools, R. (1999). Goal neglect and inhibitory limitations: Dissociable causes of interference effects in conflict situations. *Acta Psychologica*, 101(2–3), 379–394. [https://doi.org/10.1016/S0001-6918\(99\)00012-8](https://doi.org/10.1016/S0001-6918(99)00012-8)
- Devine, S., & Otto, A. R. (2022). Information about task progress modulates cognitive demand avoidance. *Cognition*, 225, 105107. <https://doi.org/10.1016/j.cognition.2022.105107>
- Dreisbach, G., & Haider, H. (2006). Preparatory adjustment of cognitive control in the task switching paradigm. *Psychonomic Bulletin & Review*, 13(2), 334–338. <https://doi.org/10.3758/bf03193853>
- Droll, J. A., & Hayhoe, M. M. (2007). Trade-offs between gaze and working memory use. *Journal of Experimental Psychology. Human Perception and Performance*, 33(6), 1352–1365. <https://doi.org/10.1037/0096-1523.33.6.1352>
- Gold, J. I., & Shadlen, M. N. (2002). Banburismus and the Brain: Decoding the Relationship between Sensory Stimuli, Decisions, and Reward. *Neuron*, 36(2), 299–308. [https://doi.org/10.1016/S0896-6273\(02\)00971-6](https://doi.org/10.1016/S0896-6273(02)00971-6)
- Goschke, T. (2000). Intentional Reconfiguration and Involuntary Persistence in Task Set

- Switching. In *Control of Cognitive Processes: Attention and Performance XVIII*.
<https://doi.org/10.7551/mitpress/1481.001.0001>
- Goschke, T. (2003). Voluntary action and cognitive control from a cognitive neuroscience perspective. In *Voluntary action: Brains, minds, and sociality* (pp. 49–85). Oxford University Press.
- Greenhouse, S. W., & Geisser, S. (1959). On methods in the analysis of profile data. *Psychometrika*, 24(2), 95–112. <https://doi.org/10.1007/BF02289823>
- Hommel, B. (2015). Between Persistence and Flexibility: The Yin and Yang of Action Control. In *Advances in motivation science* (pp. 33–67). Elsevier.
<https://doi.org/10.1016/bs.adms.2015.04.003>
- Jersild, A. T. (1927). Mental set and shift. *Archives of Psychology*, 14, 89, 81–81.
- Johnson, E., & Payne, J. (1985). *Effort and Accuracy in Choice*.
<https://doi.org/10.1287/MNSC.31.4.395>
- Kalanthroff, E., Davelaar, E. J., Henik, A., Goldfarb, L., & Usher, M. (2018). Task conflict and proactive control: A computational theory of the Stroop task. *Psychological Review*, 125(1), 59–82. <https://doi.org/10.1037/rev0000083>
- Kessler, Y., & Oberauer, K. (2014). Working memory updating latency reflects the cost of switching between maintenance and updating modes of operation. *Journal of Experimental Psychology. Learning, Memory, and Cognition*, 40(3), 738–754.
<https://doi.org/10.1037/a0035545>
- Kiesel, A., Steinhauser, M., Wendt, M., Falkenstein, M., Jost, K., Philipp, A. M., & Koch, I. (2010). Control and interference in task switching—A review. *Psychological Bulletin*, 136(5),

- 849–874. <https://doi.org/10.1037/a0019842>
- Kleinsorge, T., & Rinkenauer, G. (2012). Effects of monetary incentives on task switching. *Experimental Psychology*, 59(4), 216–226. <https://doi.org/10.1027/1618-3169/a000146>
- Kool, W., McGuire, J. T., Rosen, Z. B., & Botvinick, M. M. (2010). Decision making and the avoidance of cognitive demand. *Journal of Experimental Psychology. General*, 139(4), 665–682. <https://doi.org/10.1037/a0020198>
- Kurzban, R., Duckworth, A., Kable, J. W., & Myers, J. (2013). An opportunity cost model of subjective effort and task performance. *The Behavioral and Brain Sciences*, 36(6), 10.1017/S0140525X12003196. <https://doi.org/10.1017/S0140525X12003196>
- Liu, Y. S., Holmes, P., & Cohen, J. D. (2008). A neural network model of the Eriksen task: Reduction, analysis, and data fitting. *Neural Computation*, 20(2), 345–373. <https://doi.org/10.1162/neco.2007.08-06-313>
- Manohar, S. G., Chong, T. T.-J., Apps, M. A. J., Batla, A., Stamelou, M., Jarman, P. R., Bhatia, K. P., & Husain, M. (2015). Reward Pays the Cost of Noise Reduction in Motor and Cognitive Control. *Current Biology: CB*, 25(13), 1707–1716. <https://doi.org/10.1016/j.cub.2015.05.038>
- McGuire, J. T., & Botvinick, M. M. (2010). Prefrontal cortex, cognitive control, and the registration of decision costs. *Proceedings of the National Academy of Sciences of the United States of America*, 107(17), 7922–7926. <https://doi.org/10.1073/pnas.0910662107>
- Meiran, N. (1996). Reconfiguration of processing mode prior to task performance. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 22(6), 1423–1442.

<https://doi.org/10.1037/0278-7393.22.6.1423>

Miletić, S., Turner, B. M., Forstmann, B. U., & van Maanen, L. (2017). Parameter recovery for the Leaky Competing Accumulator model. *Journal of Mathematical Psychology*, 76, 25–50. <https://doi.org/10.1016/j.jmp.2016.12.001>

Monsell, S. (2003). Task switching. *Trends in Cognitive Sciences*, 7(3), 134–140. [https://doi.org/10.1016/S1364-6613\(03\)00028-7](https://doi.org/10.1016/S1364-6613(03)00028-7)

Musslick, S., Bizyaeva, A., Agaron, S., Leonard, N. E., & Cohen, J. D. (2019). Stability-flexibility dilemma in cognitive control: A dynamical system perspective. *Proceedings of the 41st Annual Meeting of the Cognitive Science Society*.

Musslick, S., & Cohen, J. D. (2021). Rationalizing constraints on the capacity for cognitive control. *Trends in Cognitive Sciences*, 25(9), 757–775. <https://doi.org/10.1016/j.tics.2021.06.001>

Musslick, S., Jang, S. J., Shvartsman, M., Shenhav, A., & Cohen, J. D. (2018). Constraints associated with cognitive control and the stability-flexibility dilemma. *Proceedings of the 40th Annual Meeting of the Cognitive Science Society*.

Nieuwenhuis, S., & Monsell, S. (2002). Residual costs in task switching: Testing the failure-to-engage hypothesis. *Psychonomic Bulletin & Review*, 9(1), 86–92. <https://doi.org/10.3758/BF03196259>

O'Reilly, R. C. (2006). Biologically based computational models of high-level cognition. *Science (New York, N.Y.)*, 314(5796), 91–94. <https://doi.org/10.1126/science.1127242>

Posner, M. I., & Snyder, C. R. R. (2004). *Attention and Cognitive Control* (p. 223). Psychology Press.

- Rigoux, L., Stephan, K. E., Friston, K. J., & Daunizeau, J. (2014). Bayesian model selection for group studies—Revisited. *NeuroImage*, *84*, 971–985.
<https://doi.org/10.1016/j.neuroimage.2013.08.065>
- Rogers, R. D., & Monsell, S. (1995). Costs of a predictable switch between simple cognitive tasks. *Journal of Experimental Psychology: General*, *124*(2), 207–231.
<https://doi.org/10.1037/0096-3445.124.2.207>
- Savine, A. C., Beck, S. M., Edwards, B. G., Chiew, K. S., & Braver, T. S. (2010). Enhancement of cognitive control by approach and avoidance motivational states. *Cognition & Emotion*, *24*(2), 338–356. <https://doi.org/10.1080/02699930903381564>
- Schmitz, F., & Voss, A. (2012). Decomposing task-switching costs with the diffusion model. *Journal of Experimental Psychology. Human Perception and Performance*, *38*(1), 222–250. <https://doi.org/10.1037/a0026003>
- Schmitz, F., & Voss, A. (2014). Components of task switching: A closer look at task switching and cue switching. *Acta Psychologica*, *151*, 184–196.
<https://doi.org/10.1016/j.actpsy.2014.06.009>
- Servan-Schreiber, D., Printz, H., & Cohen, J. D. (1990). A network model of catecholamine effects: Gain, signal-to-noise ratio, and behavior. *Science (New York, N.Y.)*, *249*(4971), 892–895. <https://doi.org/10.1126/science.2392679>
- Shah, A. K., & Oppenheimer, D. M. (2008). Heuristics made easy: An effort-reduction framework. *Psychological Bulletin*, *134*(2), 207–222. <https://doi.org/10.1037/0033-2909.134.2.207>
- Shenhav, A., Botvinick, M. M., & Cohen, J. D. (2013). The expected value of control: An

- integrative theory of anterior cingulate cortex function. *Neuron*, 79(2), 217–240.
<https://doi.org/10.1016/j.neuron.2013.07.007>
- Shenhav, A., Musslick, S., Lieder, F., Kool, W., Griffiths, T. L., Cohen, J. D., & Botvinick, M. M. (2017). Toward a Rational and Mechanistic Account of Mental Effort. *Annual Review of Neuroscience*, 40, 99–124. <https://doi.org/10.1146/annurev-neuro-072116-031526>
- Shvartsman, M., Srivastava, V., & Cohen, J. D. (2015). A theory of decision making under dynamic context. In C. Cortes, N. D. Lawrence, D. D. Lee, M. Sugiyama, & R. Garnett (Eds.), *Advances in neural information processing systems 28* (pp. 2485–2493). Curran Associates, Inc.
- Spector, A., & Biederman, I. (1976). Mental set and mental shift revisited. *The American Journal of Psychology*, 89(4), 669–679. <https://doi.org/10.2307/1421465>
- Srivastava, V., Feng, S. F., Cohen, J. D., Leonard, N. E., & Shenhav, A. (2017). A martingale analysis of first passage times of time-dependent Wiener diffusion models. *Journal of Mathematical Psychology*, 77, 94–110. <https://doi.org/10.1016/j.jmp.2016.10.001>
- Stephan, K. E., Penny, W. D., Daunizeau, J., Moran, R. J., & Friston, K. J. (2009). Bayesian model selection for group studies. *NeuroImage*, 46(4), 1004–1017.
<https://doi.org/10.1016/j.neuroimage.2009.03.025>
- Turner, B. M., & Sederberg, P. B. (2014). A Generalized, Likelihood-Free Method for Posterior Estimation. *Psychonomic Bulletin & Review*, 21(2), 227–250.
<https://doi.org/10.3758/s13423-013-0530-0>
- Usher, M., & McClelland, J. L. (2001). The time course of perceptual choice: The leaky, competing accumulator model. *Psychological Review*, 108(3), 550–592.

<https://doi.org/10.1037/0033-295x.108.3.550>

Van Selst, M., & Jolicoeur, P. (1994). A Solution to the Effect of Sample Size on Outlier Elimination. *The Quarterly Journal of Experimental Psychology Section A*, 47(3), 631–650. <https://doi.org/10.1080/14640749408401131>

Vandierendonck, A., Liefoghe, B., & Verbruggen, F. (2010). Task switching: Interplay of reconfiguration and interference control. *Psychological Bulletin*, 136, 601–626. <https://doi.org/10.1037/a0019791>

Wilcox, N. T. (1993). Lottery Choice: Incentives, Complexity and Decision Time. *The Economic Journal*, 103(421), 1397–1417. <https://doi.org/10.2307/2234473>

Wilcox, N. T. (1994). On a Lottery Pricing Anomaly: Time Tells the Tale. *Journal of Risk and Uncertainty*, 7(3), 311–324.

Yeung, N., & Monsell, S. (2003). Switching between tasks of unequal familiarity: The role of stimulus-attribute and response-set selection. *Journal of Experimental Psychology. Human Perception and Performance*, 29(2), 455–469. <https://doi.org/10.1037/0096-1523.29.2.455>